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High-throughput diaphragm compressor

Abstract

Devices and methods for operating a diaphragm compressor system provide high output pressure and high throughput. In some embodiments, modular diaphragm compressors are stacked with a clamping mechanism pressing the compressor modules together. In embodiments, multiple stacks are provided as stages of a pressurization process. In embodiments, a main stage valve controls one or more pressure circuits for one or more hydraulic actuators of compressor modules. In embodiments, orifices configured for damping are incorporated to control actuator piston movement within a compressor module.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/840,919 filed Jun. 15, 2022 and claims the benefit under 35 U.S.C. § 119(e) of the earlier filing date of United States Provisional Patent Application No. 63/277,125 filed on Nov. 8, 2021, the disclosure of which is incorporated by reference herein.

FIELD OF THE INVENTION

(1) The present invention is directed to diaphragm compressors and modifications for improving reliability and hydraulic efficiency in high-pressure and/or high-throughput applications.

BACKGROUND OF THE INVENTION

(2) A diaphragm compressor actuates a diaphragm at high speed to pressurize a process gas. Although some modern applications require process gas at high pressures and/or in large tanks, a conventional diaphragm compressor system is limited by physical constraints, for example the compressor head volume, speed of operation, actuation force, material strength, and the like.

SUMMARY OF THE INVENTION

(3) A feature and benefit of embodiments is a diaphragm compressor system, comprising a plurality of compressor modules mounted in a stack configuration, and a clamping mechanism configured to apply a clamping force to the first and second compressor head of each compressor module of the plurality of compressor modules. Each compressor module comprises a first compressor head, a second compressor head, and a hydraulic drive. Each of the first and second compressor heads comprises a head cavity and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region. The diaphragm is configured to actuate from a first position to a second position during a discharge cycle to pressurize process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head. The diaphragms of the first and second compressor heads of each compressor module are centered on a compressor axis. The hydraulic drive is configured to pressurize work oil and provide the pressurized work oil to the first and second compressor heads. The hydraulic drive comprises: a hydraulic power unit configured to provide a variable-pressure supply of work oil to the hydraulic drive, a plurality of pressure circuits comprising: a first pressure circuit of work oil at a first pressure, and a second pressure circuit of work oil at a second pressure, a first diaphragm piston, wherein a first variable volume region is defined between the first diaphragm piston and the diaphragm of the first compressor head, and a second diaphragm piston, wherein a second variable volume region is defined between the second diaphragm piston and the

diaphragm of the second compressor head. During a discharge cycle of a compressor head, the hydraulic power unit is configured to drive the respective diaphragm piston toward the corresponding diaphragm compressor head, intensifying the work oil in the respective variable volume region to an intensified pressure, and actuating the diaphragm to the second position. The clamping mechanism is configured to apply a clamping force to the first and second compressor head of each compressor module of the plurality of compressor modules, the clamping mechanism comprising a base plate and an end plate configured to be compressed on opposing sides of the plurality of compressor modules. The clamping mechanism is configured to increase a distance between the base plate and the end plate in response to thermal expansion of one or more compressor modules of the plurality of compressor modules, and the clamping mechanism is configured to apply the clamping force parallel to the compressor axis.

(4) In embodiments, each compressor head comprises a work oil head support plate and a process gas head support plate, wherein the clamping force of the clamping mechanism is configured to clamp together each work oil head support plate with the respective process gas head support plate for each compressor module of the plurality of compressor modules.

(5) In embodiments, the plurality of compressor modules are in a staged configuration configured to discharge process gas at a first pressure and a second pressure, and wherein the system is configured to provide the discharged process gas at the first pressure from the first compressor head of the first compressor module of the plurality of compressor modules as an inlet supply of process gas to another compressor head of the system.

(6) In embodiments, one or more of the compressor modules of the plurality of compressor modules comprises a bypass check valve configured to bypass process gas past the respective compressor module

(7) In embodiments, each compressor module comprises the first compressor head outputting process gas at a first pressure and the second compressor head outputting process gas at a second pressure, and wherein the system is configured to provide the discharged pressurized process gas from the first compressor head as an inlet supply of process gas to the second compressor head.

(8) In embodiments, the plurality of compressor modules comprises four compressor modules.

(9) In embodiments, the four compressor modules are configured to provide four sequential stages of increasing process gas pressurization.

(10) In embodiments, the four compressor modules comprise a first compressor module configured to output pressurized process gas of at least 50 bar, a second compressor module configured to output pressurized gas of at least 200 bar, a third compressor module configured to output pressurized gas of at least 600 bar, and a fourth compressor module configured to output pressurized gas of at least 800 bar.

(11) In embodiments, the clamping mechanism connects the base plate and the end plate by at least one of: at least two tie rods and a reactionary frame.

(12) In embodiments, the clamping mechanism comprises: one or more tie rods, and at least one of a plurality of pre-tensioning nuts and a plurality of Belleville spring washers. The clamping mechanism is configured to provide a pre-tension load on at least one of the base plate and the end plate.

(13) In embodiments, the clamping mechanism further comprises a clamp actuator configured to provide a dynamic clamping force to the plurality of compressor modules.

(14) In embodiments, the clamping mechanism comprises: a plurality of tie rods and a plurality of tensioner nuts.

(15) In embodiments, the hydraulic drive of each compressor module further comprising an actuator piston defining an actuator axis, wherein the actuator piston is configured to move along the actuator axis to drive the diaphragm pistons.

(16) In embodiments, the compressor axis and the actuator axis are coaxial.

(17) In embodiments, the compressor axis and the actuator axis are not coaxial.

(18) In embodiments, each compressor module of the plurality of compressor modules is configured to be selectively deactivated, wherein, when a compressor module of the plurality of compressor modules is deactivated, the compressor system is configured to operate the remaining compressor modules of the plurality of compressor modules.

(19) In embodiments, the hydraulic drive of each compressor module comprises: the first pressure circuit comprising a low-pressure circuit, the second pressure circuit comprising a medium-pressure circuit, and a third pressure circuit comprising a high-pressure circuit of work oil at a third pressure, and the medium-pressure circuit comprising a first main stage valve and the high-pressure circuit comprising a second main stage valve, each main stage valve configured to control the flow of work oil to or from the hydraulic drive, and each main stage valve configured to control the flow of work oil to selectively drive at least two compressor heads of the compressor system.

(20) In embodiments, the hydraulic drive of each compressor module further comprising an actuator piston configured to drive the diaphragm pistons, and the first main stage valve configured to control a flow of the medium-pressure circuit to or from either side of the actuator piston, and the second main stage valve configured to control a flow of high-pressure work oil to either side of the actuator piston.

(21) In embodiments, the hydraulic power unit is configured to supply work oil from one or more pressure circuits of the plurality of pressure circuits to each hydraulic drive of two or more compressor modules of the plurality of compressor modules.

(22) A feature and benefit of embodiments is a method of pressurizing a process gas, comprising: providing the diaphragm compressor system of above as a first stage; providing a second of the diaphragm compressor system of above as a second stage, wherein the second stage is configured to provide a higher maximum outlet pressure than the first stage; operating a low-pressure efficient mode, filling the tank with pressurized process gas from each of the first and second stages; and operating a high-pressure mode with the first and second stages in series. The low-pressure efficient mode, comprises: shutting off or bypassing the second stage, supplying a process gas at an inlet pressure to the first stage, filling a tank with pressurized process gas from the first stage. The operating a high-throughput mode with the first and second stages in parallel comprises: supplying a process gas at an inlet pressure to each of the first and second stages, and filling the tank with pressurized process gas from each of the first and second stages. The operating a high-pressure mode with the first and second stages in series comprises: supplying a process gas at an inlet pressure to the first stage, supplying pressurized process gas from the first stage to the second stage.

(23) A feature and benefit of embodiments is a compressor system, comprising: a plurality of compressor modules mounted in a stack configuration, a clamping mechanism configured to apply a clamping force to the compressor head of each compressor module of the plurality of compressor modules, and a tank configured to retain pressurized process gas that is discharged from one or more of the compressor modules of the plurality of compressor modules. Each compressor module comprises a compressor head and a hydraulic drive. The compressor head comprises a head cavity and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region. The diaphragm is configured to: actuate from a first position to a second position during a discharge cycle to pressurize process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head. The hydraulic drive is configured to pressurize work oil and provide the pressurized work oil to the compressor head. The hydraulic drive comprises: a hydraulic power unit configured to provide a variable-pressure supply of work oil to the hydraulic drive, and a plurality of pressure circuits comprising: a first pressure circuit of work oil at a first pressure, and a second pressure circuit of work oil at a second pressure. During a discharge cycle of a compressor head, the variable-pressure supply of work oil is configured to drive the respective diaphragm piston toward the corresponding diaphragm compressor head, intensifying the work oil in the variable

volume region to an intensified pressure, and actuating the diaphragm to the second position. The clamping mechanism comprises: a base plate and an end plate being compressed on opposing sides of the plurality of compressor modules, and a clamp actuator configured to provide a dynamic clamping force to the plurality of compressor modules. The plurality of compressor modules comprises four compressor modules configured to provide four sequential stages of increasing process gas pressurization.

(24) A feature and benefit of embodiments is a compressor system, comprising: a plurality of compressor modules mounted in a stack configuration and a clamping mechanism. Each compressor module comprises a first compressor head, a second compressor head, a hydraulic power unit, and a plurality of pressure circuits. Each of the first and second compressor heads comprising: a head cavity, and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region. The diaphragm is configured to: actuate from a first position to a second position during a discharge cycle to pressurize process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head. The hydraulic power unit is configured to provide a variable-pressure supply of work oil to the first and second compressor heads. The plurality of pressure circuits comprising: a first pressure circuit of work oil at a first pressure, and a second pressure circuit of work oil at a second pressure. During a discharge cycle of a compressor head, the variable-pressure supply of work oil is configured to drive the respective diaphragm toward the corresponding process gas region, intensifying the work oil in the work oil region to an intensified pressure, and actuating the diaphragm to the second position. The clamping mechanism is configured to apply a clamping force to the first and second compressor head of each compressor module of the plurality of compressor modules.

(25) The above summary of the various representative embodiments of the invention is not intended to describe each illustrated embodiment or every implementation of the invention. Rather, the embodiments are chosen and described so that others skilled in the art can appreciate and understand the principles and practices of the invention. The Figures in the detailed description that follow more particularly exemplify these embodiments.

(26) Another feature and benefit of embodiments is a diaphragm compressor system, comprising a first compressor head and a hydraulic drive. The first compressor head comprises a head cavity and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region. The diaphragm is configured to actuate from a first position to a second position during a discharge cycle to pressurize process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head. The hydraulic drive is configured to pressurize work oil and provide the pressurized work oil to the compressor head. The hydraulic drive comprises a drive housing and an actuator piston. The drive housing comprises a drive cavity, a plurality of ports, a first plurality of orifices in communication with both the drive cavity and one or more ports of the plurality of ports, a second plurality of orifices in communication with both the drive cavity and one or more ports of the plurality of ports, and a piston subassembly. The plurality of ports comprises a first distal port and a second distal port, wherein the hydraulic drive is configured to provide a variable-pressure supply of work oil to the drive cavity through one or more of the plurality of ports. The piston subassembly comprises a first diaphragm piston mounted in the drive cavity and comprising a first diameter and an actuator piston located in the drive cavity. A first variable volume region comprises the work oil region of the compressor head and is defined between the first diaphragm piston and the diaphragm of the first compressor head. The actuator piston is located in the drive cavity, the actuator piston dividing the drive cavity into a first actuation volume in communication with the first distal port and the first plurality of orifices and a second actuation volume in communication with the second distal port and the second plurality of orifices. The actuator piston comprises a first side oriented toward the first actuation volume and a second side oriented toward the second

actuation volume. During the discharge cycle of the first compressor head: the hydraulic drive is configured to provide the variable-pressure supply of work oil through the second port to the second actuation volume to press against the second side of the actuator piston to drive the actuator piston, driving the first diaphragm piston to actuate the corresponding first compressor head, intensifying the work oil in the first variable volume region to an intensified pressure, and actuating the diaphragm of the first compressor head to the second position, the drive cavity is configured to dampen the driving of the actuator piston due to a volume of work oil in the first actuation volume that vents through the first plurality of orifices during driving of the actuator piston, and the first plurality of orifices is configured to be open to the first actuation volume when the driving of the actuator piston begins, and the plurality of orifices being progressively covered by the actuator piston during the driving, increasing a damping force of work oil remaining in the first actuation volume against the first side of the actuator piston.

(27) In embodiments, the drive housing further comprises a plurality of supplemental orifices in communication with the first actuation volume, the plurality of supplemental orifices being staggered axially relative to the plurality of first orifices, the plurality of supplemental orifices configured to dampen the driving of the actuator piston due to the volume of work oil in the first actuation volume that vents through the plurality of supplemental orifices during driving of the actuator piston.

(28) In embodiments, each of the plurality of supplemental orifices comprise a smaller area than the each of the first plurality of orifices.

(29) In embodiments, the plurality of supplemental orifices are configured to be progressively covered by the actuator piston during the driving, increasing the damping force of work oil remaining in the first actuation volume against the first side of the actuator piston.

(30) In embodiments, the actuator piston is configured to drive in a linear direction along an actuator piston axis, and the first plurality of orifices is formed in one or more surfaces of the drive housing oriented at a non-parallel angle relative to the actuator piston axis.

(31) In embodiments, the plurality of first orifices extend radially away from the drive cavity and the actuator piston.

(32) In embodiments, the diaphragm compressor system further comprises one or more plugs installed in one or more of the first plurality of orifices to increase the damping force of the first actuation volume.

(33) In embodiments, an annular gap is between the actuator piston and the drive housing, the annular gap configured to dampen the driving of the actuator piston by controlling an outflow of work oil in the first actuation volume during the discharge stroke.

(34) In embodiments, the annular gap is configured to dampen the driving of the actuator piston after the first plurality of orifices are obstructed by the actuator piston.

(35) In embodiments, the actuator piston comprises a first internal porting and a first opening in the first side, wherein the first distal port is selectively in fluid communication with the first actuation volume via the first internal porting.

(36) In embodiments, during the discharge cycle of the first compressor head, the work oil in the first actuation volume also vents through the first internal porting of the actuator piston during driving of the actuator piston, wherein the first distal port is configured to receive both the vented work oil from the first plurality of orifices and the vented work oil from the internal porting, and wherein the hydraulic drive is configured to fill an accumulator of the low-pressure circuit from the vented work oil in the first distal port.

(37) In embodiments, a landing orifice operatively connects the first distal port and a first proximal port, the landing orifice configured to control the flowrate of work oil venting through the first internal porting.

(38) In embodiments, a feedback mechanism is configured to determine at least one of a position and velocity of the actuator piston during use.

(39) In embodiments, the actuator piston comprises a variable-geometry portion, and the feedback mechanism comprising a sensor configured to detect a distance from the variable-geometry portion during the driving of the actuator piston.

(40) In embodiments, the feedback mechanism comprises a pressure sensor operatively coupled to process gas in the first compressor head or discharged therefrom, the feedback mechanism configured to determine the velocity of the actuator piston based on the pressure of the discharged process gas.

(41) In embodiments, the first actuation volume is in direct communication with a first proximal port, and the second actuation volume in direct communication with a second proximal port.

(42) In embodiments, the drive housing further comprises a removable sleeve insert, wherein the sleeve insert comprises a first distal annulus in fluid communication between the first actuation volume and the first distal port and a second distal annulus in fluid communication between the second actuation volume and the second distal port.

(43) In embodiments, the sleeve insert further comprises: a first proximal annulus in fluid communication between the first actuation volume and the first proximal port; and a landing orifice operatively connecting the first distal annulus and a first proximal annulus, the landing orifice configured to control the flowrate of work oil venting through the first internal porting.

(44) In embodiments, the first plurality of orifices comprises twelve or more orifices, and wherein the second plurality of orifices comprises twelve or more orifices.

(45) In embodiments, the orifices are in communication with at least one of the ports of the plurality of ports, and the orifices are configured to: during a stroke of the actuator piston toward the corresponding first or second actuation volume of the drive cavity, vent work oil from the same corresponding first or second actuation volume to the at least one of the ports, and during a stroke of the actuator piston in the opposite direction, supply work oil from the at least one of the ports to the drive cavity to pressurize the same corresponding first or second actuation volume.

(46) In embodiments, the diaphragm compressor system further comprises a second compressor head and a second diaphragm piston, wherein, during the discharge cycle of the second compressor head: the hydraulic drive is configured to provide the variable-pressure supply of work oil through the first distal port to the first actuation volume to press against the first side of the actuator piston to drive the actuator piston, driving the second diaphragm piston to actuate the corresponding second compressor head, intensifying the work oil in a second variable volume region to an intensified pressure, and actuating a diaphragm of the second compressor head, the drive housing is configured to dampen the driving of the actuator piston due to a volume of work oil in the second actuation volume that vents through the second plurality of orifices during driving of the actuator piston, and the second plurality of orifices is configured to be open to the second actuation volume when the driving of the actuator piston begins, and the plurality of orifices being progressively covered by the actuator piston during the driving, increasing a damping force of work oil remaining in the second actuation volume against the second side of the actuator piston.

(47) Still another feature and benefit of embodiments is a diaphragm compressor system with main stage valve controlling a hydraulic drive thereof, the diaphragm compressor comprising: a main stage valve and a diaphragm compressor system. The main stage valve comprises: a valve body comprising a first end and a second end, a pilot port proximate the first end, a supply port, a first vent port, a cylinder port, and a pin subassembly comprising a spool, a pilot pin proximate the second end, and a return pin proximate the first end. The pin subassembly is configured to move to a supply position with the supply port in fluid communication with the cylinder port and an end orifice configured to dampen motion of the pin subassembly into the supply position. The pin subassembly is configured to move to a vent position with the cylinder port in fluid communication with the first vent port and a vent damping orifice configured to dampen motion of the pin subassembly into the vent position. The diaphragm compressor system comprises: a first compressor head and a hydraulic drive. The first compressor head comprises: a head cavity, and a

diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region. The diaphragm is configured to actuate during a discharge cycle to pressurize process gas in the process gas region. The hydraulic drive is configured to pressurize work oil and provide the pressurized work oil to the compressor head, the hydraulic drive comprising: a drive housing, a hydraulic power unit, and a plurality of pressure circuits. The drive housing comprises a plurality of ports and a drive cavity. The plurality of ports comprise a first port and a second port, wherein the hydraulic drive is configured to provide a variable-pressure supply of work oil to the drive cavity through one or more of the plurality of ports. The drive cavity is divided into a first actuation volume in communication with the first port and a second actuation volume in communication with the second port. The plurality of pressure circuits comprise a low-pressure circuit of work oil at a low pressure, and a high-pressure circuit of work oil at a high pressure. During the discharge cycle of the first compressor head: the hydraulic drive is configured to provide the variable-pressure supply of work oil through the second port to the second actuation volume **146**, intensifying the work oil in the first variable volume region to an intensified pressure, and actuating the diaphragm of the first compressor head to the second position. The main stage valve is mounted to the drive housing with the cylinder port in fluid communication with the drive cavity and the supply port operatively coupled to the hydraulic power unit, and wherein the main stage valve is configured to selectively move to the supply position to connect the high-pressure circuit to the drive cavity during the discharge cycle of the first compressor head.

(48) In embodiments, during a suction cycle of the first compressor head: the pin subassembly is configured to move to the vent position to connect the drive cavity of the drive housing to the first vent port of the main stage valve, and the hydraulic drive vents work oil from the second actuation volume through the main stage valve.

(49) In embodiments, the low-pressure circuit of the hydraulic drive further comprises a recovered oil accumulator operatively coupled to the first vent port of the main stage valve, the main stage valve configured to supply oil from the drive cavity to the recovered oil accumulator when in the vent position.

(50) In embodiments, the low-pressure circuit comprises the recovered oil accumulator.

(51) In embodiments, the diaphragm compressor system with main stage valve of claim **4**, further comprising a passive valve operatively connected to the recovered oil accumulator and the drive cavity. During the suction cycle of the first compressor head, the passive valve is configured to supply oil from the recovered oil accumulator to the drive cavity.

(52) In embodiments, the diaphragm compressor system further comprises: a multi-stage pilot valve mounted in the second end of the valve body, the pilot valve configured to selectively actuate the pin subassembly of the main stage valve. The hydraulic drive further comprises a pilot pressure circuit and a pilot pressure accumulator operatively coupled to the pilot valve.

(53) In embodiments, the pilot pressure circuit is operatively coupled to the pilot port at the first end of the valve body, and the pin subassembly has a larger area proximate the pilot valve than proximate the pilot port and the pin subassembly is configured to move to the supply position when pilot pressure is supplied to the pilot pin through the pilot valve and the pilot port.

(54) In embodiments, a return spring is configured to bias the pin subassembly toward the vent position when pressure is not supplied to the pilot valve.

(55) In embodiments, the main stage supply valve is a first main stage valve operatively coupled to the first actuation volume of the drive cavity, the diaphragm compressor system with main stage valve further comprises a second main stage valve. The second main stage valve is mounted to the drive housing with each of the vent port and the cylinder port in fluid communication with the second actuation volume of the drive cavity and the supply port operatively coupled to the hydraulic power unit. The second main stage valve is configured to selectively move to the supply position to connect the high-pressure circuit to the second actuation volume.

(56) In embodiments, at least one of an actuator and a valve mounted in the second end of the valve

body is configured to selectively actuate the pin subassembly of the main stage valve to the supply position.

(57) In embodiments, one or more of the pilot port and the vent damping orifice comprises a plurality of rows of damping orifices comprising a row of relatively larger damping orifices proximate the spool and a row of smaller damping orifices proximate one of the respective first and second end.

(58) In embodiments, the pilot port comprises a ring of removable orifices.

(59) In embodiments, a diaphragm compressor system comprises: a plurality of pressure circuits comprising a medium-pressure circuit and a high-pressure circuit and four or more of the main stage valve described above. The system comprises a first main stage valve configured to selectively supply the high-pressure circuit to the first actuation volume and vent work oil from the first actuation volume, a second main stage valve configured to selectively supply the high-pressure circuit to the second actuation volume and vent work oil from the second actuation volume, a third main stage valve configured to selectively supply the medium-pressure circuit to the first actuation volume and vent work oil from the first actuation volume, and a fourth main stage valve configured to selectively supply the medium-pressure circuit to the second actuation volume and vent work oil from the second actuation volume.

(60) In embodiments, the first and second main stage valves are each configured to vent from the respective first or second actuation volume through the respective first vent port to a recovered oil accumulator.

(61) In embodiments, the diaphragm compressor system further comprises a passive valve configured to supply oil from the recovered oil accumulator to the drive cavity during the suction cycle of the first compressor head.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention can be completely understood in consideration of the following detailed description of various embodiments of the invention in connection with the accompanying drawings, in which:

(2) FIG. 1 is a front perspective view of a high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(3) FIG. 2 is a top front perspective view of an embodiment of a compressor module of the system of FIG. 1 in accord with embodiments of the present disclosure.

(4) FIG. 3 is a bottom rear perspective view of the compressor module of FIG. 2.

(5) FIG. 4 is front elevation view of the compressor module of FIG. 2.

(6) FIG. 5 is side elevation view of the compressor module of FIG. 2.

(7) FIG. 6 is top sectional view of the compressor module of FIG. 2.

(8) FIG. 7 is side sectional view of the compressor module of FIG. 2.

(9) FIG. 8 is an enlarged partial view of FIG. 7.

(10) FIG. 9 is a side sectional view of another embodiment of a compressor module of the system of FIG. 1 in accord with embodiments of the present disclosure.

(11) FIG. 10 is a side sectional view of still another embodiment of a compressor module of the system of FIG. 1 in accord with embodiments of the present disclosure.

(12) FIG. 11 is an enlarged partial view of FIG. 10.

(13) FIG. 12 is a sectional view of a compressor head for a compressor module in accord with embodiments of the present disclosure.

(14) FIG. 13 is a top perspective wireframe view of another compressor head for a compressor module in accord with embodiments of the present disclosure.

(15) FIG. **14** is a schematic view of a hydraulically-driven compressor module with two compressor heads force coupled in accord with embodiments of the present disclosure.

(16) FIG. **15** is a hydraulic circuit diagram of a hydraulically-driven compressor module with three pressure rails in accord with embodiments of the present disclosure.

(17) FIG. **16** is a schematic view of a hydraulically-driven compressor module with three pressure rails in accord with embodiments of the present disclosure.

(18) FIG. **17** is a hydraulic circuit diagram of a hydraulically-driven compressor module with three pressure rails in accord with embodiments of the present disclosure.

(19) FIG. **18A** is a cross-sectional view of a main stage valve of the high-throughput compressor system in accord with embodiments of the present disclosure in a vent position.

(20) FIG. **18B** is a cross-sectional view of the main stage valve of FIG. **18A** in a supply position.

(21) FIG. **19** is a partial cross-sectional view of a compressor module of the system of FIG. **1** in accord with embodiments of the present disclosure.

(22) FIG. **20** is a top perspective wireframe view of a valve manifold of the compressor module of FIG. **2** in accord with embodiments of the present disclosure.

(23) FIG. **21** is a top sectional view of a hydraulic clamp actuator of the system of FIG. **1** in accord with embodiments of the present disclosure.

(24) FIG. **22** is a hydraulic circuit diagram of a staged arrangement of multiple stacks of the high-throughput compressor system of FIG. **1** in accord with embodiments of the present disclosure.

(25) FIG. **23** is a hydraulic circuit diagram of another staged arrangement of multiple stacks of the high-throughput compressor system of FIG. **1** in accord with embodiments of the present disclosure.

(26) FIG. **24** is a front perspective view of another high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(27) FIG. **25** is a front perspective view of still another high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(28) FIG. **26** is a front perspective view of yet another high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(29) FIG. **27** is a cross-sectional view of the system of FIG. **26**.

(30) FIG. **28** is a front perspective view of another high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(31) FIG. **29** is a front perspective view of still another high-throughput compressor system with stacked compressor modules in accord with embodiments of the present disclosure.

(32) FIG. **30** is a schematic view of multiple hydraulically-driven compressor modules with a common intensifier in accord with embodiments of the present disclosure.

(33) FIG. **31** is a schematic view of multiple hydraulically-driven compressor modules with a common control valve in accord with embodiments of the present disclosure.

(34) FIG. **32** is a schematic view of a hydraulically-driven compressor system with direct hydraulic actuation in accord with embodiments of the present disclosure.

(35) FIG. **33** is a schematic view of a hydraulically-driven compressor module with an active oil injection system in accord with embodiments of the present disclosure.

(36) While the invention is amenable to various modifications and alternative forms, specifics thereof have been depicted by way of example in the drawings and will be described in detail. It should be understood, however, that the intention is not to limit the invention to the particular embodiments described. On the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

(37) As shown in FIG. **1**, in embodiments of the present disclosure, a high-throughput compressor system **200** comprises multiple compressor modules **100**, for example compressor modules **100A**,

100B, 100C, 100D, collectively referred to as a stack **201** of compressor modules **100**. Each compressor module **100** is a diaphragm compressor with one or more compressor heads **31, 51** each having a diaphragm **5**.

(38) The process gas may be any gas suitable for pressurization for any use. In embodiments, the process gas is hydrogen. For embodiments designed for filling stations for hydrogen fuel cell vehicles, the required outlet pressure of the high-throughput compressor system **200** may be approximately 10,000-12,000 psi. In embodiments, the target pressure of stored hydrogen in a tank (e.g., at a vehicle filling station) is up to about 14,500 psi to account for pressure losses in, e.g., storage and transfer. Therefore, the corresponding discharge pressure of the process gas from the high-throughput compressor system **200** in such embodiments is about 15,000 psi.

(39) Diaphragm Compressor

(40) Applicable embodiments of the architecture and function of an individual diaphragm compressor **1** are shown in FIGS. **12** and **13** and may be similar to the compressor disclosed in U.S. patent application Ser. No. 17/522,896, the entire contents of which are incorporated herein by reference and for all purposes. Relative to the present disclosure, the compressor **1** in U.S. Ser. No. 17/522,896 constitutes an embodiment of each diaphragm compressor head **31, 51**, for example the diaphragm compressor heads of the compressor module **100**. Similar diaphragm compressors and related systems are also disclosed in U.S. Provisional Application Nos. 63/111,356 filed Nov. 9, 2020 and 63/277,125 filed on Nov. 8, 2021, and U.S. patent application Ser. No. 17/522,896 filed Nov. 9, 2021, the entire contents of which are incorporated herein by reference and for all purposes.

(41) In embodiments, the diaphragm compressor **1** is driven by a diaphragm piston **3** (also referred to as a high-pressure oil piston) that moves a volume of work oil (i.e., hydraulic fluid) through the compressor **1** suction and discharge cycles. Process gas compression occurs as the volume of work oil is pushed towards the diaphragm **5** by a diaphragm piston **3** to fill a work oil region **35** in a work oil head support plate **8** (or lower plate or oil plate), exerting a uniform force against the bottom of the diaphragm **5**. This deflects the diaphragm **5** into an upper cavity in a gas plate **6** that is filled with the process gas, also referred to as a process gas region **36**. The deflection of the diaphragm **5** against the upper cavity of gas plate **6** first compresses the process gas and then expels it through an outlet port **9** comprising a discharge check valve. As the oil piston **3** reverses to begin the suction cycle, the diaphragm **5** is drawn downward towards the oil plate **8** while the inlet check valve at the inlet port **7** opens and fills the process gas region **36** with a fresh charge of process gas at an inlet pressure. The diaphragm piston **3** reaches the end of its stroke before beginning its next stroke, and the compression cycle is repeated.

(42) In embodiments, the compressor head **31** comprises a process gas head support plate **6**, a work oil head support plate **8**, and a diaphragm **5**. The process gas head support plate **6** comprises a process gas inlet port **7** operatively connected to an inlet check valve and a process gas outlet port **9** operatively connected to a discharge check valve. In certain embodiments, the work oil head support plate **8** comprises an inlet **33** operatively connected to one or more inlet check valves **45**, and an outlet **34** operatively connected to one or more relief valves **42** (inlet check valves and relief valves shown schematically in FIG. **33**). A head cavity **15** is defined between the process gas head support plate **6** and the work oil head support plate **8**. In certain embodiments, the compressor head **31** comprises a piston bore **32** extending toward the work oil head support plate **8** and sized to receive the diaphragm piston **3**. In other embodiments, there is no piston bore **32** and the diaphragm piston **3** is configured to remain substantially within the drive housing **114**.

(43) The diaphragm **5** is mounted in the head cavity **15** between the process gas head support plate **6** and the work oil head support plate **8** and divides the head cavity into a work oil region **35** and a process gas region **36**. The diaphragm piston **3** defines the volume of the work oil region **35** between a top face of the diaphragm piston **3** and a bottom face of the diaphragm **5**. Because the diaphragm piston **3** and diaphragm **5** are dynamic, the volume of the work oil region **35** is variable.

(44) The diaphragm **5** is configured to actuate from a first position proximate the work oil head

support plate **8** (e.g., in contact with the work oil head support plate or fully extended toward the work oil head support plate) to a second position proximate the process gas head support plate **6** during a discharge cycle to pressurize process gas in the process gas region **36** from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the outlet port **9**. During a suction cycle of the compressor head **31**, the diaphragm **5** is configured to move from the second position to the first position to fill the process gas region **36** with process gas at the inlet pressure. In embodiments, the diaphragm **5** is a diaphragm set comprising a plurality of diaphragm plates sandwiched together and acting in unison, for example two, three, four, or more diaphragm plates may comprise a diaphragm set. In certain embodiments, the diaphragm plates are made from a metal. In other embodiments, the diaphragm plates are made from different metals. In other embodiments, one or more of the diaphragm plates are not made from metal. In certain embodiments, the diaphragm **5** includes three plates, the three plates comprising stainless steel in the outside plates, and brass on the inside plate.

(45) Compressor heads applicable to embodiments of the present disclosure may be provided in any of various sizes and compression ratios. In embodiments, an individual compressor head **31** may be configured for a pressure range of process gas outlet of 200 psi to 15,000 psi. In other embodiments, a compressor head **31** may be configured for a maximum pressure range of 40 psi to 30,000 psi. In still further embodiments, a compressor head **31** may be configured for a pressure range of 300 psi to 45,000 psi. In certain embodiments, the aforementioned compressor heads **31** may be run at pressures below 200 psi, 40 psi, and 300 psi, respectively. In some embodiments, a compressor head **31** can have a compression ratio range of 0.25:1, 1:1, 2:1, 3:1, 4:1, 5:1, 6:1, 10:1, 20:1, and ranges therebetween.

(46) Hydraulically-Driven Compressor Modules

(47) Referring to FIGS. 2-8, an embodiment of a compressor module **100** is shown. Applicable embodiments of the architecture and function of the individual compressor module **100** are discussed in U.S. patent application Ser. No. 17/522,896 (therein referred to as a “compressor system”). The compressor module **100** comprises a first compressor head **31** and a second compressor head **51**. The compressor module **100** in some embodiments is hydraulically driven by a hydraulic drive **110** that is configured to intensify or pressurize work oil and provide the intensified work oil to the first and second compressor heads **31**, **51**. In embodiments, the hydraulic drive **110** comprises an actuator **112**, a drive housing **114** defining a drive cavity **116**, and a hydraulic power unit **118** (“HPU”) providing pressurized hydraulic fluid at a pressure, which effectively supplies a pressurized circuit **120** (also referred to broadly as a pressure rail, volume of work oil at a given pressure, or flow of work oil at a given pressure). In embodiments, the hydraulic drive **110** includes one or more pressurized circuits **120** provided by one or more HPUs **118**, and in further embodiments, the actuator **112** comprises a piston subassembly **122**. In some embodiments, the hydraulic drive **110** is configured to provide a variable-pressure supply of work oil to the drive cavity **116** from one or more of: different pressures of work oil in a one or more pressurized circuits **120**, variable areas of components of the piston subassembly **122** (e.g., a variable-area architecture), and/or variable control of the piston subassembly.

(48) In certain embodiments, the piston subassembly **122** (e.g. as shown in FIG. 6) comprises the diaphragm piston **3** mounted at least partially in the drive housing **114** and extending into the piston bore **32** (FIG. 6, see also FIG. 12). In some embodiments, the piston bore **32** is formed partially or completely in the drive housing **114** (e.g., FIG. 6). A first variable volume region **54** comprises the work oil region **35** of the compressor head **31** along with the available volume of the piston bore **32**; in other words, the first variable volume region **54** is defined between the diaphragm piston **3** and the diaphragm **5** of the corresponding compressor head **31**. The piston subassembly **122** comprises an actuator piston **126** located in the drive cavity **116** and coupled (directly or indirectly, for example rigidly coupled, mechanically linked, or hydraulically coupled) to the diaphragm piston **3**, the actuator piston defining an actuator piston axis **208**. The diaphragm piston **3** is

coupled to the actuator piston **126** to move in response to movement of the actuator piston **126**. In some embodiments, the diaphragm piston **3** is mechanically rigidly fixed to the actuator piston **126** or, as shown in FIGS. **6-10**, formed as a unitary one-piece part with the actuator piston. In other words, the diaphragm piston **3** may be one control area and the actuator piston **126** another control area of the same unitary piston; likewise, a second diaphragm piston **140** (discussed below) may be a third control area.

(49) FIGS. **2-8, 9**, and **10-11** illustrate embodiments of a compressor module **100** applicable to the present disclosure that is dual-headed and comprises the compressor head **31** and the second compressor head **51**. FIGS. **14-17** schematically illustrate embodiments of the hydraulic drive **110** for a dual-headed compressor module **100**, although the hydraulic drive is applicable to a compressor module having any number of heads, for example 1-6 heads. The second compressor head **51** is actuated by a second diaphragm piston **140** defining a second variable volume region **142**. In some embodiments, the piston subassembly **122** is mounted in the drive cavity **116** of the drive housing **114** and a plurality of variable volumes are provided between the piston subassembly **122** and the drive housing **114**.

(50) As shown in FIG. **8**, a first actuation volume **144** is defined on the side of the actuator piston **126** proximate the compressor head **31**, and a second actuation volume **146** is defined on the opposite side of the actuator piston and proximate the second compressor head **51**. Other embodiments may include one, three, or more variable volumes. Due to movement of the piston subassembly **122**, the first and second actuation volumes **144, 146** are variable in volume and defined as the volume between the respective first and second sides **143, 145** of the actuator piston **126** and the interior of the respective first and second diaphragm pistons **3, 140**. The variable volume is a result of the movement of the actuator piston **126** back and forth. As discussed below, in certain operating states, the first and second actuation volumes **144, 146** also serve a damping function against the actuator piston **126** as it is being driven.

(51) Referring to FIGS. **6-11**, the drive housing **114** also comprises a plurality of ports **147** in communication with the first and second actuation volumes **144, 146**. In embodiments, the ports **147** include a first distal port **148A** for the first actuation volume **144** and a second distal port **148B** for the second actuation volume **146**. The hydraulic drive **110** is operatively connected to one or more of these actuator volumes **144, 146** through one or more of the plurality of ports **147**. The hydraulic drive **110** is configured to supply work oil or vent work oil as required by the operating conditions of the compressor module **100**. In some embodiments, one or more main stage valves **250** (“MSV **250**”) control the flow of work oil to or from one or more of these ports **147** and thereby control the flow of work oil to or from a respective actuation volume **144, 146** (see, e.g., FIG. **14**). It will be appreciated that in embodiments any one or more of the of the plurality of ports **147** may be a plurality of ports arranged around the actuator piston **126**, for example the cross-sectional view of FIG. **7** illustrates two each (at top and bottom) of the first and second distal ports **148A, 148B** along with two each of the first and second proximal ports **148C, 148D**. In certain embodiments, the plurality of first and second distal ports **148A, 148B** are arranged around the actuator piston **126**. The plurality of first and second distal ports **148A, 148B** may be arranged annularly and/or symmetrically about the actuator piston axis **208**. In embodiments, the drive housing **114** comprises one or more manifold ports **117** for connecting the plurality of ports **147** to exterior components (e.g., a valve manifold **244** discussed below with reference to FIG. **20**).

(52) As shown in FIG. **16**, in some embodiments, four MSVs **250A-D** are provided as two for each of the first and second actuation volumes **144, 146**, each MSV corresponding to a pressurized circuit of the one or more pressurized circuits **120**. In this embodiment, for the first actuation volume **144**, the MSV **250C** controls a medium-pressure circuit **132** and the MSV **250A** controls a high-pressure circuit **134**; for the second actuation volume **146**, the MSV **250D** controls the medium-pressure circuit **132** and the MSV **250B** controls the high-pressure circuit **134**. In another sense of this embodiment, each pressure circuit comprises two MSVs **250**, one for supplying work

oil and one for venting work oil during a piston stroke, with those roles reversed during the opposite stroke. For example, during the discharge stroke of compressor head **31**, MSV **250B** provides a supply of high pressure work oil through the high-pressure circuit **134**, while MSV **250C** vents work oil on the other side of the actuator piston **126** out of the first actuation volume **144**.

(53) As shown in FIGS. **2-8**, in embodiments, the compressor module **100** comprises a first diaphragm compressor head **31** and a second diaphragm compressor head **51** that are each aligned and centered on a compressor axis **206** extending through the center of the diaphragm **5**. In certain embodiments, the first diaphragm compressor head **31** and the second diaphragm compressor head **51** are driven by a single hydraulic actuator **114**. In some embodiments, the hydraulic actuator **114** is operatively coupled to both the first and second diaphragm compressor heads **31**, **51**, such that the suction cycle of one compressor head aids in initiating the discharge cycle of the other compressor head, which creates a force couple between the compressor heads as discussed further below.

(54) In certain embodiments, the process gas discharged from a compressor head (e.g., first compressor head **31**) is at a relatively low pressure and, for further pressurization, may subsequently be fed into another compressor head, which may be either the second compressor head **51** of the same compressor module **100**, or a compressor head **31**, **51** of a separate compressor module **100B-D** of the same stack **201**, or a compressor head **31**, **51** of a compressor module of a separate stack for further compression.

(55) In some embodiments, the compressor module **100** is arranged compactly and therefore requires specific hydraulic routing and high pressure gas plumbing and connections. In some embodiments, the compressor heads **31**, **51** can accommodate reorientation of the inlet and outlet ports **7**, **9**. As shown in FIG. **2**, the 180° opposing inlet port **7** and outlet port **9** can be clocked in almost any desired orientation as indicated by the arrows **A**.

(56) In certain embodiments, an energy recovery mechanism can be provided through a force couple architecture, embodiments of which are shown in FIGS. **2-8**, **9**, and **10-11**. Referring to FIGS. **2-8**, some embodiments of this architecture comprise a pair of opposing diaphragm compressor heads **31**, **51** both driven by an actuator piston **126** that is a double acting double rod, which may or may not act as a hydraulic intensifier, and which is actuated to provide high pressure work oil to actuate the diaphragm compressors. The two pressurized actuation volumes **144**, **146** are alternately fed pressurized fluid and vented to drive the actuator piston **126** back and forth towards either compressor head **31**, **51**. Additionally and as discussed further below, since the respective diaphragms **5** of the compressor heads **31**, **51** oppose each other and are out of phase in this embodiment, the force imposed on one diaphragm by the intake of process gas (e.g., intake of process gas to compressor head **31**) consequently imposes an aiding force during the opposing diaphragm's compression and discharge stroke (e.g., compression and discharge from compressor head **51**). The force couple architecture imposes a force couple to the actuator **114** reducing the force and energy requirements for moving the actuator piston **126** to actuate the diaphragms **5** of both compressor heads **31**, **51**.

(57) For a discharge cycle of the compressor head **31**, operation begins when the actuator piston **126** is at or near the end of its stroke away from the compressor head **31**. At this point, process gas at the inlet pressure has already been supplied to the process gas region **36** of the diaphragm compressor head **31** whereas the opposing second compressor head **51** is fully evacuated of process gas. When diaphragm **5** motion is desired for the compressor head **31**, the MSV **250** actuates to supply pressurized work oil to the second actuation volume **146** on the second side **145** of the actuator piston **126**, forcing the actuator piston **126** up towards the compressor head **31** that is filled with process gas ("up" and other such directions are in reference to FIG. **6** for sake of clarity and are an example embodiment of the relative movement and positions of various parts, but are not intended to be limiting). As the actuator piston **126** moves, the diaphragm piston **3** pressurizes the

work oil in the work oil region **54** below the diaphragm **5**. Since this hydraulic pressure in the work oil region **54** is greater than the pressure of process gas, the diaphragm **5** moves upwards thereby pressurizing the process gas. Once the process gas pressure reaches a target process gas pressure, the process gas is expelled out of the compressor head **31** and either supplied to the tank **256** or supplied to a subsequent compressor head (e.g., the compressor head **51** of the same compressor module **100**, a compressor head **31**, **51** of another compressor module in the same stack **201**, or a compressor head **31**, **51** of another compressor module in another stack) for further pressurization. After all or most of the process gas has been forced out of the process gas region **36**, the MSV **250** stops providing hydraulic flow and the actuator piston **126** stops actuating.

(58) When diaphragm motion is desired in the opposing direction (i.e., a discharge stroke of the second compressor head **51**), the MSV **250** is actuated to provide pressure to the opposing first side **143** of the actuator piston **126** into the first actuation volume **144**, thereby forcing the actuator piston in the opposite direction and compressing the gas in the second variable volume region **142** toward the second compressor head **51**. As the hydraulic actuator **112** pressurizes the process gas within the second compressor head **51**, the compressor head **31** is undergoing its intake or suction stroke where the process gas at inlet pressure is supplied above the diaphragm **5** in the process gas region **36**. This initial supply of inlet-pressure process gas may initially assist in providing pressure and moving the diaphragm **5** downwardly and pressurizes the remaining work oil below the diaphragm **5** in the variable volume region **54**, which applies a force to the diaphragm piston **3** thereby providing an aiding force during the opposing compressor head **51** compression, or discharge stroke. This aiding force from the process gas supply reduces the required force from the HPU **118** to drive the actuator piston **140** and compress gas in the second compressor head **51**. Subsequently, process gas completely fills the compressor head **31** in process gas region **36**. To finish the discharge stroke of the second compressor head **51**, pressurized process gas is discharged from the process gas region **36** of the second compressor head **51**. Upon completion of the discharge stroke of the second compressor head **51**, the compressor head **31** is filled with process gas and the second compressor head **51** is fully evacuated of process gas.

(59) Referring to FIG. **9**, another embodiment of a compressor module **100** includes a first and second internal porting **127A**, **127B** through the respective first and second sides **143**, **145** of the actuator piston **126**. The first side **143** of the actuator piston **126** comprises a first opening **154** and the first internal porting **127A** that are in fluid communication with both the first actuation volume **144** and the first proximal port **148C**. In this manner, the first actuation volume **144** can be supplied or vented through the first internal porting **127A** and the first opening **154**. In the illustrated embodiment, the first proximal port **148C** is part of a low-pressure circuit **130** and is controlled by a main stage valve **250** ("MSV **250**") to selectively supply low-pressure work oil to the first actuation volume **144** or vent work oil from the first actuation volume. In other embodiments, the first internal porting **127A** may be in fluid communication with any one or more of the plurality of ports **147** and operable with any one or more of a plurality of pressurized circuits **120**.

(60) In embodiments, at least one of the first opening **154** and the first internal porting **127A** of the actuator piston **126** comprises a check valve (not shown) to prevent the flow of work oil out of the first actuation volume **144** through the first internal porting **127A** when the first actuation volume is pressurized for a discharge stroke of the second diaphragm piston **140**, the check valve thereby maintaining the pressure in the first actuation volume **144**. As discussed below, in some embodiments a landing orifice **107** connects the first proximal port **148C** to the first distal port **148A**, and vented work oil from the first internal porting **127A** flows out through the first distal port **148A** via the landing orifice to a pressurized circuit, accumulator, or the reservoir **230**. In some embodiments, additional ports (e.g., first proximal port **148C** in FIG. **9**) of the plurality of ports **147** are in fluid communication with the first actuation volume **144** separately from or in addition to the first opening **154**. It will be appreciated that, in embodiments, the first opening **154** and the second internal porting **127B** are provided at the second side **145** of the actuator piston **126** and in

communication with the second actuation volume **146** in a substantially similar manner as at the first side **143** of the actuator piston.

(61) Referring to FIGS. **10-11**, still another embodiment of a compressor module **100** is shown that is generally similar to FIG. **9**. In embodiments, the first and second compressor heads **31, 51** comprise an oil distribution plate **55** including an array of passages **56** from the respective variable volume region **54, 142** to the diaphragm **5**.

(62) Referring to FIG. **10**, in some embodiments, the compressor module **100** comprises a feedback mechanism **108** configured to determine one or more of a position and velocity of the actuator piston **126** during use. The feedback mechanism may include one or more of a sensor **158** and a pressure sensor **159**. In some embodiments, the actuator piston **126** comprises an indication feature **156** that is detectable by the sensor **158**. In various embodiments, the sensor **158** is one or more of an inductive sensor, an optical sensor, a Hall Effect sensor, or the like.

(63) In certain embodiments, the indication feature **156** is a variable-geometry portion of the actuator piston **126**, for example a decreasing radius, and the sensor **158** is an inductive proximity sensor configured to measure the distance to the indication feature **158**, the distance measured in a direction perpendicular to the motion of the actuator piston **126** along the actuator piston axis **208**. In one such embodiment shown in FIG. **10**, as the actuator piston **126** moves right-to-left toward the first compressor head **31**, the sensor **158** can detect an increase in the distance to the indication feature **158** because the radius of the actuator piston is decreasing. Based on the measured distance between the sensor **158** and the indication feature **158**, the feedback mechanism **108** is configured to determine the absolute position of the actuator piston **126**. In embodiments, the feedback mechanism **108** is configured to determine the velocity of the actuator piston **126** based on multiple measurements by the sensor **158** over time.

(64) In embodiments, the feedback mechanism **108** comprises a pressure sensor **159** (FIG. **15**) operatively coupled to pressurized process gas in or from the compressor head **31**, for example directly measuring pressure of the process gas in the process gas region **36** or measuring the pressure of discharged process gas from the first compressor head **31** after the inlet port **7**. The feedback mechanism **108** is configured to calculate the velocity of the actuator piston **126** based on the measured pressure of the discharged process gas. In embodiments, the feedback mechanism **108** is configured to calculate the velocity of the actuator piston based on multiple inputs, such as measurement(s) from the sensor **158** in conjunction with the pressure sensor **159** or other sensors operatively configured to sense or detect a portion of the compressor module **100** and/or hydraulic drive **110** (e.g., pressure sensor(s) in the first and/or second variable volume region **54, 142**, pressure sensor(s) in the first and/or second actuation volume **144, 146**). The feedback mechanism **108** is configured to control other aspects of the module **100** based on the position and/or velocity of the actuator piston **126**, for example controlling the main stage valves **250** to supply or vent work oil to the hydraulic drive **110** or controlling the supply of process gas to a compressor head **31, 51**.

(65) Referring to FIG. **19**, an alternative embodiment of a compressor module **100** is shown with the hydraulic drive **110** positioned offset from one or more compressor heads **31, 51** with only a hydraulic passage manifold **114B** between the compressor heads. The hydraulic passage manifold **114B** provides passages that hydraulically connects the hydraulic drive **110** to the compressor heads **31, 51** without pistons and is significantly smaller than the drive housing **114** of other embodiments. This arrangement reduces the axial length of the stack **201** and may reduce the overall footprint of the stack **201** or the entire high-throughput compressor system **200**. In the illustrated embodiment, the compressor head axis **206** is perpendicular to the actuator piston axis **208**. However, compressor head axis **206** can be oriented in nearly any relationship to the actuator piston axis **208** so long as they are in fluid communication.

(66) In embodiments, the compressor heads **31, 51** of the compressor module **100** may be independently operated and timed to be synchronized, not synchronized, or alternating. Such

arrangements are generally achievable in any compressor architecture that is not force coupled. In embodiments, the compressor heads **31, 51** are discharged at substantially the same time. Similarly in embodiments of a stack **201** of compressor modules **100** or a stage **202** of compressor modules **100**, the timing of discharge cycles for compressor heads **31, 51** may be independent or dependent within each module, stack, or stage. In embodiments providing independent operation of the compressor heads **31, 51**, one or more actuator pistons **126** are separately provided for each compressor head. In certain such embodiments, one or more ports of the plurality of ports **147** are dedicated to a given individual compressor head **31, 51** for control of the respective compressor head. In any of the above embodiments with independent operation, any one or more compressor head **31, 51**, compressor module **100**, or stack **201** may be selectively turned off and on during operation of the diaphragm compressor system **200**, for example turned off when not needed during certain stages of filling the tank **256**.

(67) Pressure Rails

(68) The hydraulic system pressure(s) provided by the hydraulic power unit **118** (“HPU”) in some embodiments ranges from 0-5000 psi, but in other embodiments a higher hydraulic pressure is implemented. The HPU **118** in embodiments comprises a single pump/motor, many small pump/motor systems, or fewer larger pump/motor systems, or combinations thereof, as based on operational requirements. In embodiments, the hydraulic drive **110** comprises actively-controlled pressure-compensated pumps or the like in order to actively control hydraulic pressure throughout operating modes. This active control enables the hydraulic drive **110** to operate efficiently by minimizing energy expenditure to meet system requirements. The HPU **118** is configured to provide work oil at a pressure to the drive cavity **116**, and in some embodiments, this pressure is intensified, e.g., by increasing the supply area relative to the piston area.

(69) For some embodiments, in order to minimize hydraulic energy consumption, a variable pressure architecture of the compressor module **100** provides a variable-pressure supply of work oil to provide step or analog changes in the applied pressure to any actuator piston **126** as discussed in U.S. patent application Ser. No. 17/522,896. Accordingly, in embodiments, for different operating modes, the hydraulic drive **110** may supply work oil at multiple different set pressures (also referred to as pressure circuits **120** or pressure rails) and/or flowrates. The plurality of pressure circuits **120** comprises one or more low-pressure circuits **130**, medium-pressure circuits **132**, and high-pressure circuits **134**. The term “circuit” is intended to broadly include both the pressurized fluid and the associated structures conveying and controlling the fluid, and one skilled in the art will appreciate that the same structure (e.g., plumbing) may serve as a part of multiple circuits depending on the operating conditions.

(70) In some embodiments with multiple pressure circuits, the HPU **118** uses discrete pump/motor sets producing discrete pressures that supply some or all of the plurality of pressure circuits **120** individually in order to eliminate throttling losses. In embodiments, the HPU **118** comprises a variable pump-motor set that is configured to change the speed or pressure of pressurized work oil output by the HPU. In embodiments, the HPU **118** is automatically variable and/or actively controlled, for example, controlled and adjusted in response to conditions in the hydraulic drive **110**, conditions of the outlet process gas or conditions in the fill tank **256**. Moreover, in certain embodiments, any of the above approaches is used to charge one or more accumulators **136** that are included in one or more of the plurality of pressure circuits **120**.

(71) In embodiments of the variable pressure architecture, a low-pressure circuit **130** is implemented to provide a “backfill” or “assist” hydraulic supply to the hydraulic system **100** when a higher pressure is not needed (e.g., when ambient-pressure work oil or other relatively low-pressure work oil is sufficient). In certain embodiments, as the hydraulic actuator starts to move from the end of its stroke, the force imposed by the intake stroke process gas on the diaphragm **5** imposes an aiding force on the diaphragm piston **3** and consequently on the actuator **112**. In some embodiments, this force may be enough to move the actuator **112**, or initiate movement of the

actuator **112**, with minimal pressure from the HPU **118** or without the addition of hydraulic pressure to available work oil. The drive cavity **116**, however, will still need a supply of work oil to backfill in one of the actuation volumes **144**, **146** to allow the actuator **112** to move in the opposite direction, which may be provided by the low-pressure supply rail **130**. In embodiments, the low-pressure circuit **130** comprises relatively low-pressure work oil from one or more of the following: unpressurized work oil from the HPU **118**, an oil reservoir **38** of an active oil injection system **30** providing a circuit of supplemental work oil to the compressor heads, vented work oil from the drive cavity **116** in a previous cycle (e.g., intensified work oil vented via a valve and stored in a hydraulic accumulator **136D** as discussed below), vented work oil from the variable volume region **54**, process gas at the inlet pressure, or other sources in the compressor system **100**.

(72) In certain embodiments, the one or more pressure circuits **120** comprises a medium-pressure circuit **132** comprising work oil pressurized by the HPU **118** (e.g., by a throttled supply of higher pressure work oil or by a direct supply from one or more pumps/motors of the HPU). In some embodiments, the one or more pressure circuits **120** comprises a high-pressure circuit **134** comprising high-pressure work oil pressurized by the HPU **118**. It will be appreciated that any of the low-pressure circuit **130**, medium-pressure circuit **132**, and high-pressure circuit **134** may be implemented as multiple pressure circuits at different set pressures. The additional circuits of the plurality of pressure circuits **120** allow for finer tuning and control of the compressor module **100**, and increases efficiency by only providing as much pressure as necessary to move the actuator at a particular part of its stroke.

(73) As discussed above, in embodiments the compressor module **100** is configured to control the variable-pressure supply of work oil by supplying work oil from the high-pressure circuit **134** after work oil has been supplied from the low-pressure circuit **130** and/or the medium-pressure circuit **132**. In certain embodiments, the hydraulic drive system **110** is configured to control the variable-pressure supply of work oil by sequentially providing work oil to the drive cavity **116** from the low-pressure circuit **130**, the medium-pressure circuit **132**, and the high-pressure circuit **134**. In embodiments with low pressure operating conditions or requirements, it may be sufficient to provide the work oil to the drive cavity **116** from the low-pressure circuit **130** and the medium-pressure circuit **132**, only.

(74) In some embodiments, the plurality of pressure circuits **120** are each operatively connected to the drive cavity **116** and may be fed on one or both sides of the actuator piston **126**. In embodiments, the hydraulic drive **110** comprises a passive first valve **131** (FIG. **17**) configured to supply work oil from the low-pressure rail of the low-pressure circuit **130** to the drive cavity **116** and an active second valve **133** (FIG. **16**) configured to supply work oil from the medium-pressure rail of the medium-pressure circuit **132** to the drive cavity. Certain embodiments further comprise an active third valve **135** configured to supply work oil from the high-pressure rail of the high-pressure circuit **134** to the drive cavity **116**. As detailed below, in embodiments the active second valve **133** and/or the active third valve **135** may be the main stage valve **250** (“MSV”).

(75) In certain embodiments, each of the active second valve **133** and the active third valve **135** is configured to adjust from a supply stage to a return stage, the return stage permitting an outflow of intensified work oil from the drive cavity **116** during the discharge cycle of a corresponding one of the compressor heads **31**, **51**. In embodiments, a hydraulic accumulator **136D** receives the outflow of intensified work oil from the drive cavity **116**. The hydraulic accumulator **136D** in some embodiments operatively functions as a low-pressure accumulator **136A** of the low-pressure circuit **130**, a medium-pressure accumulator **136B** of the medium-pressure circuit **132**, a high-pressure accumulator **136C** of the high-pressure circuit **134**, or a pilot accumulator **136E** of the pilot valve **290**.

(76) Supplying flow from the low-pressure circuit **130** into the hydraulic actuator **112** can be achieved several ways. In some embodiments, the fluid can be supplied through a hydraulic valve (in place of the passive first valve **131** in FIG. **17**) that opens to allow flow into the actuator **112**

then closes when higher pressure fluid is required. In other embodiments, the flow can be supplied through a check valve, such as the passive first valve **131** which opens due to low pressure of work oil in the work oil region **35** during a suction cycle and as the hydraulic actuator **112** starts to move. Since this is a passive valve, it does not need to be actuated when relatively higher pressure fluid is supplied to the drive cavity **116** (e.g., from the medium-pressure circuit **132** or the high-pressure circuit **134**) will force the valve closed. Alternately, a three-way valve can be used to supply low-pressure or high-pressure fluid to the hydraulic actuator **112** and vent from the hydraulic actuator when desired. The vent can be connected to the low-pressure circuit **130** as outlined above. In this scenario, fluid from the low-pressure circuit **130** can back flow through the passive first valve **131** into the hydraulic actuator **112** as the actuator starts to move.

(77) In certain embodiments, a medium-pressure circuit **132** is set to a pressure approximately 50% of the high pressure circuit **134**. In other embodiments, a medium-pressure circuit **132** is set to a pressure approximately 40% to 60% of the high pressure circuit **134**. In some embodiments, the high-pressure circuit **134** is set at a pressure of approximately 5,000 psi, the medium-pressure circuit **132** is set to from 2,500 psi to 3,000 psi, and the low pressure circuit **130** is set to approximately 500 psi. In other embodiments, high-pressure circuit **134** is set to a pressure selected from 3,000 psi, 5,000 psi, and 7,500 psi. In some embodiments, at least one of the high-pressure circuit **134** and medium-pressure circuit **32** are controlled by the HPU **118** to be variable from the maximum pressure for each respective rail. In other embodiments, at least one of the high-pressure circuit **134** and medium-pressure circuits **132** are controlled by the HPU to be variable in a range from 0% to 100% of the maximum pressure for each respective rail. In further embodiments, at least one of the high-pressure circuit **134** and medium-pressure circuits **132** are controlled by the HPU to be variable in a range from 50% to 100% of the maximum pressure for each respective rail.

(78) In certain embodiments, the compressor module **100** may include two stages, for example a low pressure stage with compressor head **31** and a high pressure stage with second compressor head **51**, as discussed in U.S. patent application Ser. No. 17/522,896.

(79) FIGS. **30-32** show alternative embodiments where some components of the hydraulic drive **110** are shared over multiple compressor heads **31**, **51** that may be part of a stack **201** applicable to the present disclosure.

(80) Referring to FIG. **30**, in embodiments a common actuator **240** is operatively coupled to several compressor heads, for example compressor heads **31A-D**, while being physically offset from the compressor heads. This arrangement is in contrast to other embodiments with one actuator for each compressor module that is physically housed in the module between compressor heads. The common actuator **240** functions as the intensifier and hydraulic drive for each compressor head **31A-D**. The common actuator **240** in embodiments is driven by a single HPU **118** or multiple HPUs. In certain embodiments and as illustrated in FIG. **30**, the common actuator **240** provides pressurized fluid to simultaneously actuate the diaphragms **5** of both compressor heads **31A**, **31B**, and then the common actuator **240** reverses directions to simultaneously actuate the diaphragms **5** of both compressor heads **31C**, **31D**.

(81) In some embodiments of the stack **201**, the common actuator **240** is mounted in the stack with the actuator piston axis **208** coaxial with the compressor head axis **206**. In other embodiments, the common actuator **240** is separate from the stack **201**. In still other embodiments, the common actuator **240** and the corresponding compressor heads **31A-D** are all separate from any stack **201** to operate as an auxiliary compressor plumbed to another compressor module or stack. In any such embodiments, a single compressor head of the compressor heads **31A-D** may be configured to be taken offline while the remaining compressor heads continue to operate.

(82) Referring to FIG. **31**, in some embodiments a first and second compressor module **100A**, **100B** share a common control valve, MSV **250**. The MSV controls a pressurized supply of work oil from the HPU **118**. In this sense, the HPU **118** and the MSV **250** are configured to supply and control the supply of pressurized work oil to a plurality of compressor modules and a plurality of

compressor heads. In other embodiments, multiple MSVs **250** and/or multiple HPUs **118** are provided and shared by the first and second compressor modules **100A**, **100B**, for example when providing both medium-pressure and high-pressure circuits **132**, **134**.

(83) In an embodiment shown in FIG. **32**, the HPU **118** is configured to act directly on the diaphragm **5** of one or more compressors **31** while omitting hydraulic actuator **112** and the piston subassembly **122**. The MSV **250** is operatively connected to the HPU **118** to control the supply of work oil directly to the diaphragms **5**. In the illustrated embodiment, the MSV **250** controls the supply to three compressors **31**. In embodiments, any one or more of the pressure circuits **120** is implemented and controlled by one or more MSVs **250** for one or more of the compressor heads **31**. Although FIG. **32** is illustrated schematically, it will be appreciated that the physical arrangement of the compressor heads **31** of the stack **201** can be coaxial on a compressor axis **206** as in previous embodiments. A compressor stack **201** implementing this embodiment provides an axial length and overall footprint are significantly decreased.

(84) In any embodiments of the present disclosure, each pressure circuit of the one or more pressure circuits **120** may be independently and actively controlled to adjust the amount of pressure supplied to the hydraulic actuator **112**. In embodiments, the active valves **133**, **135** or MSV **250** may be controlled to adjust the respective pressure circuit. The plurality of ports **147** may similarly comprise a valve to actively control or throttle the flow to the drive cavity **116**. It will be appreciated that the hydraulic drive **110** is likewise configured for nearly instantaneous stoppage of the actuator piston **126** and shutdown of the compressor module **100** due to the HPU **118** along with the active valves **133**, **135** and/or the MSV **250**, any associated control mechanisms (e.g., feedback mechanism **108**), or shutoff valves. For example, the actuator piston **126** can be stopped during a discharge or suction stroke before such stroke is completed by closing off the pressure circuit(s) that are pressurizing the corresponding actuation volume. Accordingly, the hydraulic drive **110** is configured to stop a stroke of the actuator piston **126**, a stroke of the diaphragm piston(s) **3**, **140**, and/or actuation of the diaphragm(s) **5** before the stroke or actuation completes its current cycle. This shutoff capability provides safety by minimizing further damage when a hazardous condition is detected. This is an improvement over prior compressor drives, e.g., crank-driven systems, which must mechanically stop components and overcome significant inertial forces before stopping, resulting in continuing operation during the hazardous condition.

(85) Clamping Mechanism

(86) Referring to FIG. **1**, in embodiments, the high-throughput compressor system **200** comprises a clamping mechanism **204** that holds the compressor modules **100** together while accommodating the significant pressures, vibrations, and other forces experienced during compressor cycles. In other embodiments, a clamping mechanism **304**, **404**, **504**, **604**, or **704** is provided as discussed below, with broad functionality similar to the clamping mechanism **204** (see FIGS. **24-29**). Generally, each individual compressor head **31**, **51** is formed of multiple plates that must be clamped together with enough force to resist cyclical forces including pressurized work oil, pressurized process gas, and diaphragm actuation without leakage. As such, a conventional individual compressor head requires a specialized individual mechanism such as a large number of high-strength bolts to sufficiently clamp the head together. By contrast, the clamping mechanism **204** of the present disclosure applies a clamping force sufficient to hold together each such compressor head for multiple compressor modules **100** with minimal or no clamping within individual heads.

(87) In certain embodiments, the clamping force is exerted by the clamping mechanism **204** at opposite ends of a stack **201** of one or more compressor modules **100**, with the force acting through each module **100** to clamp all of the heads **31**, **51** of each module **100** in the stack. Clamping together each head **31**, **51** comprises clamping together support plates that define each head, resisting pressure of compressed fluid(s) inside the head, and clamping one or more of the support plates (e.g., work oil head support plate **8**) to a drive housing **114** of the compressor module **100**. It

will be appreciated that the clamping mechanism **204** therefore eliminates and/or reduces other hardware, including bolts and also the size and thickness of components of the heads **31, 51**, necessary for clamping a conventional compressor head, and may provide reduced assembly time, reduced size and weight for each module **100** compared to a conventional diaphragm compressor, and improved serviceability. The clamping mechanism **204** is therefore also applicable to a single compressor head **31, 51** or a single compressor module **100** that is not stacked with other modules. In certain embodiments, the total clamping force necessary to operate each head **31, 51** in a stack **201** of modules **100** is not provided by bolts securing each individual head **31, 51** to each respective module **100**. In other embodiments, the total clamping force is provided by a combination of the clamping mechanism **204** and bolts securing each individual head **31, 51**.

(88) In embodiments, the compressor system **200** comprises a clamp actuator **212** that applies a compressive force (i.e., clamping load) to the compressor modules **100** while also accommodating changes in thermal expansion of the compressor modules during operation. If the stack **201** were rigidly clamped without the clamp actuator **212**, significant stresses would arise in the compressor modules **100** due to thermal growth of hardware that results from temperature increases as process gas is compressed. While accommodating thermal expansion, the compressive force of the clamp actuator **212** may be constant or substantially constant. In embodiments, the clamp actuator **212** is a hydraulic load actuator. The present disclosure provides several embodiments that are based around an actuator such as the clamp actuator **212** and a reactionary structure such as a frame or tie rod arrangement.

(89) In the embodiment of FIG. **1**, the clamping mechanism **204** comprises a base plate **220** and an end plate **222** connected by four tie rods **224** with respective tensioner nuts **226**. In some embodiments, the stack **201** is mounted on a skid **228** or similar base. The compressor modules **100** are aligned along a common axis which, as discussed above, is a compressor head axis **206** of each compressor module **100**. In some embodiments, one or both of the base plate **220** and end plate **224** are movable or repositionable along the compressor axis **206** to accommodate thermal expansion or various sizes of compressor modules **100**. In certain embodiments, one or both of the base plate **220** and the end plate **224** are movable by the clamping actuator **212**.

(90) Referring also to FIG. **21**, the clamp actuator **212** in embodiments is a hydraulically-powered piston actuator comprising a hydraulic cylinder **232** and a piston **233**. In the illustrated embodiment, the hydraulic cylinder **232** would have pressure applied to create the required clamping force for the constituent compressor modules **100** and compressor heads **31, 51**. The same clamp actuator **212** architecture could be used for all compressor head sizes, but with different clamping loads as required for different supply pressures. In embodiments, the hydraulic supply for the clamp actuator **212** may be shared with one or more of the compressor modules **100**, for example from a hydraulic power unit **118** of a compressor module.

(91) In embodiments, operation of the clamp actuator **212** comprises manually charging the hydraulic cylinder **232** and subsequently monitoring the pressure within the clamp actuator **212**. The cylinder **232** is then resupplied if pressure drops and leakage occurs over time, for example through dynamic seals. Alternatively, the hydraulic circuit of the clamp actuator **212** could be supplied with an additional make-up pump (not shown) to accommodate the lost fluid. The make-up pump may be similar to, or the same as used for the active oil injection system **30** applicable to embodiments of the compressor module **100** discussed below, and would be an adequate option to provide a low flow high pressure supply source. In embodiments, the monitoring may be automated with a configuration where the compressor system **200** shuts down when the pressure within the clamp actuator **212** drops below a certain threshold, and oil may also be injected, or oil pressure increased, as detected by the system.

(92) In certain embodiments, as one mode to accommodate thermal growth, the large volume of oil within the hydraulic cylinder **232** of the clamp actuator **212** inherently has some compressibility, which may account for some of the increased pressure from modules **100** due to thermal growth.

Even with this compressibility, the clamp load could grow by nearly 25% with a four-module stack **201**. This load increase is proportional to the length of the overall stack **201**. To further reduce the stiffness of the volume, some embodiments incorporate a piston accumulator (not shown), for example a high pressure piston accumulator that can accommodate pressures up to 1,000 bar. Moreover, such an accumulator also allows additional time for system shut down if a seal were to fail. In some embodiments, the pressure is monitored, and a lower threshold is set such that when the pressure drops below the set point, the system shuts down and sends an alarm to the operator. The lower set point, however, is still within a reasonable clamp load such that the compressor modules remain under load during the shutdown. Since the compressor modules are hydraulically driven, the shutdown may be rapid.

(93) For service and change out of compressor modules **100**, once any plumbing or electrical connections are disconnected from a given compressor module **100**, the individual compressor module can be deactivated, serviced, and/or removed as a unit. In embodiments, an overhead gantry crane (not shown) is integrated into the stack skid **228**, and has a designated area at the end or off to the side. In embodiments, each compressor module **100** comprises an eye bolt **246** (FIG. 2) or other attachment for lifting and moving the module via the gantry crane or the like. In some embodiments, each compressor module **100** comprises one or more feet **242**, such as four feet shown in FIG. 3, which may function as a cart to move the module and/or may function to engage the skid **228** or other base of the stack **201**.

(94) Referring to FIG. 24, in other embodiments, a clamping mechanism **304** comprises a base plate **320** and end plate **322** that are connected by two tie rods **324**. The clamp actuator **212** may similarly be provided between the base plate **320** and the compressor modules **100**. Both of these embodiments of the tie rod based clamping mechanisms **204**, **304** have their own respective advantages. The four-bolt arrangement of clamping mechanism **204** can allow for easier service (e.g., service in place, change out, or temporary removal for service) of the compressor modules **100** but may be more difficult to accommodate plumbing. The two-bolt arrangement of clamping mechanism **304** provides easier plumbing access but may be more difficult to service a module.

(95) Referring to FIG. 25, in still other embodiments, a clamping mechanism **404** comprises a base plate **420**, an end plate **422**, and a reactionary frame **425** mounting the plates along with the compressor stack **201** and a clamp actuator **412**. In some embodiments, the actuator **412** may be the same as the clamp actuator **212**. In embodiments, the reactionary frame **425** is rigidly affixed (for example, bolted) to a foundation. In some embodiments, to withstand the large tensile loads applied, the reactionary frame **425** is a one-piece unitary component or two pieces rigidly affixed together, and in particular embodiments the reactionary frame is formed of one cast metal part or two cast metal parts rigidly affixed together. As shown in FIG. 26, this embodiment of clamping mechanism **404** leaves the top and sides of the compressor modules **100** substantially open and accessible, which provides access for plumbing and servicing. However, the overall size of the clamping mechanism **404** may be larger than other embodiments.

(96) Referring to FIGS. 26-27, a clamping mechanism **504** provides a different embodiment based on the four tie rod embodiment. The clamping mechanism **504** comprises a base plate **520** and end plate **522** that are connected by four tie rods **524**, and a clamp actuator **512** mounted to the base plate. In this embodiment, pre-tensioning nuts **526** are mounted on the tie rods **524** at the base plate **520** and apply an initial preload to the stack **201** before operation. The remainder of the required clamping force is then made up by the clamp actuator **512**. As the clamping actuator **512** pressure is applied, a thermal expansion gap **513** is created at the base of the actuator to accommodate thermal expansion.

(97) One benefit of this embodiment is that the pre-tensioned of the tie rods **524** creates a safety if the clamp actuator **512** fails such that the stack **201** is contained to the initial preload. However, service of each compressor module **100** may be more challenging than some of the other embodiments since the tie rods **524** are mounted more closely to the compressor modules.

(98) Referring to FIG. 28, an embodiment of the clamping mechanism **604** provides a different embodiment based on the four tie rod embodiment. The clamping mechanism **604** comprises a base plate **620** and end plate **622** that are connected by four tie rod assemblies, each including a first tie rod **624A** and a second tie rod **624B** connected by a coupler **623**. In embodiments, an additional plate **621** is mounted inside the base plate **620**. In this embodiment, the coupler **623** provides for ease of assembly and service by separately receiving the first and second tie rods **624A**, **624B**, which allows for the first and second tie rods to be relatively shorter and able to be installed from each respective side of the stack instead of assembling one long tie rod across the entire stack. In embodiments, the coupler **623** is a threaded collar. In other embodiments, the coupler **623** may provide some or all of the clamping load and thermal accommodation. In the illustrated embodiment, the tie rods **624A-B** are entirely outside of the modules **100**, providing easier access for lifting the modules out of the stack.

(99) Referring to FIG. 29, an embodiment of the clamping mechanism **704** provides a variation on the four tie rod concept. The clamping mechanism **704** comprises a base plate **720** and end plate **722** that are connected by four tie rods **724**. In this embodiment, Belleville washers **730** are mounted on the tie rods **724** at the end plate **722** and to apply a clamping load to the stack **201**. The Belleville washers are springs that are mounted to be biased in the direction of the clamping force. In use, the Belleville washers **730** receive the axial load from thermal expansion and may compress under this load while maintaining the requisite clamping force against the stack **201**. For stacks **201** of a different number of size of compressor modules **100**, the clamping mechanism **704** may be adjusted by selecting Belleville washers **730** of different sizes or a different number stacked. Additional clamping force is provided by the clamp actuator **712**.

(100) Another embodiment of the present disclosure incorporates a hydraulic actuator (such as clamp actuator **212**) to apply initial load and a threaded lock ring (not illustrated) to mechanically maintain the load. This eliminates piston seal failure as a failure mode. However, this approach provides limited thermal accommodation, or requires an adjustable lock ring.

(101) In other embodiments, the clamp actuator **212** is modified to minimize axial length, reducing the overall footprint of the stack **201**. If the parameters of the high throughput compressor system **200** and the compressor modules **100** are known, the length of the piston **233** and the cylinder **232** can be decreased to a minimum size that is capable of providing the corresponding required reactive clamping force and thermal accommodation. Additional concepts may be employed to reduce overall piston size, such as introducing a lever (e.g., a single lever arm or compound lever) functionally between the piston and the stack **201**, for example with one end of the lever rigidly affixed to the piston and the other end of the lever rigidly affixed to the second compressor head **51** of the first compressor module **100A**. The fulcrum of the lever may be mounted to the skid **228** or incorporated with other fixed parts of the clamping mechanism **204/304/404/504/604/704**. Such a lever arm multiplies the linear force of a downsized piston.

(102) Staging and Reconfigurability

(103) Embodiments of the high throughput compressor system **200** provide several potential arrangements of interconnecting and controlling the plurality of compressor modules **100** to customize a tank-filling operation. In various embodiments, some or all of the compressor modules **100** may be operatively arranged in parallel to pressurize a larger volume of process gas than would be accomplished by a single compressor module **100** over a period of time. In some embodiments, some or all of the compressor modules **100** may be operatively arranged in series to progressively pressurize process gas to a higher pressure than would be accomplished by a single compressor module **100** or a single constituent compressor head. In certain embodiments, the high throughput compressor system **200** is controlled and configurable to switch between such modes, or to combine such modes, e.g. some modules **100** operating in series and some modules **100** operating in parallel. For example, many or all of the diaphragms **5** can be used in parallel to start filling a tank **256** at low pressure. In this embodiment, some compressor heads **31**, **51** may be low pressure

heads **31**, and some may be high pressure heads **51**. In this mode, both high **51** and low **31** pressure heads are used to pressurize process gas to the pressure of the low pressure head **31** or less. Then as the pressure rises the system can be reconfigured to be a two-stage compressor (i.e., first and second stages of different pressure in series) by switching valves. In this embodiment, the low pressure head **31** pressurizes gas from a low pressure to a medium pressure, and this medium pressure gas is fed to the high pressure head **51** that pressurizes the process gas from a medium pressure to a high pressure. The heads **31**, **51** can be configured such that the compressor system **200** can have as many stages as are desired for optimizing the tank fill process.

(104) In embodiments, the compressor modules **100** are mechanically driven, e.g. with a cam driven shaft. In other embodiments, the compressor modules **100** are hydraulically driven. In order to provide high throughput and high pressure of process gas under the physical constraints of hydraulic power and diaphragm actuation, the compressor modules **100** of the present disclosure are configured to operate at high speeds with precision to prevent damage to any components. To this end, embodiments of the compressor modules **100** implement fast valving, particularly in a main stage valve **250** (“MSV **250**”) controlling the hydraulic supply, and damping of moving components such as pistons or valves.

(105) In embodiments, one or more compressor modules **100** may have compressor heads **31**, **51** that are different from the heads of other compressor module(s) with regard to pressure discharge, throughput, and/or size. In some non-illustrated embodiments, a compressor module **100** may have compressor heads **31**, **51** that are different from each other, or a different number of compressor heads such as one, three, four, or more compressor heads.

(106) Referring to FIG. **22**, an embodiment of the high-throughput compressor system **200** includes two stages **202A** and **202B**. In certain embodiments, components of the compressor system **200** are configured for process gas inlet at 120 bar. In some embodiments, each stage **202A-B** is split into two stacks **201A-B** and **201 C-D** respectively; each stack **201A-D** comprising four compressor modules **100A-D**, eight compressor heads **31**, **51**, and four hydraulic drives **110**. In certain embodiments, each hydraulic drive **110** has a dedicated HPU **118** with a single pump/motor combination or a dedicated pump/motor group. This arrangement is advantageous from an operational flexibility perspective, in that an individual hydraulic drive **110** (serving two compressor heads) can be taken offline, and its corresponding HPU **118** pump(s)/motor(s) turned off completely, while the rest of the compressor system **200** continues to operate. In one embodiment of a two-stage, single pressure supply scenario, each HPU **118** comprises a 250 hp motor and a pump sized at either 270 or 360 cc/rev. The flow requirements may be satisfied with a 2” ID supply rail hose, and return rails can be joined into a single large, welded pipe.

(107) Referring to FIG. **23**, another embodiment of the high-throughput compressor system **200** is illustrated with four stages **202A-D** of increasing pressure output and stage bypassing to allow reconfigurability. In some embodiments, three bypass valves **257** are implemented respectively preceding the second, third, and fourth stages **202B-D** to selectively bypass these latter stages. With stage bypassing, when the tank pressure is low, only the lower pressure stages may be doing the compression work, and in one embodiment the discharge process gas may bypass the upper stages. This embodiment avoids pressure drop through the additional lines, check valves and intercoolers of the upper stages, improving efficiency. In embodiments, the bypass valves **257** are three-way valves or an equivalent combination of 2-way valves. It will be appreciated that the compressor system **200** may operate when the tank **256** pressure is at any level within the operating parameters, i.e., from 0-100% of the target tank fill pressure. Accordingly, the compressor system may begin filling a higher pressure tank **256** with the high pressure stages.

(108) In the illustrated embodiment of FIG. **23**, the first stage **202A** is the lowest pressure stage and comprises six compressor heads **31**, **51** arranged as three compressor modules **100A-C**, with each compressor head configured to pressurize process gas to 1,000 psi. The second stage **202B** is the second lowest pressure stage and comprises three compressor modules **100A-C** totaling six

compressor heads **31, 51**, with each compressor head configured to pressurize process gas to 2,000 psi. The first and second stages **202A, 202B** may be in separate respective first and second stacks **201A, 201B** or arranged in a single stick **201A-B**. The third stage **202C** is the second highest pressure stage and comprises sixteen compressor heads **31, 51** arranged as eight compressor modules **100A-H** in one or two stacks **201C**, with each compressor head configured to pressurize process gas to 7,500 psi. The fourth stage **202D** is the highest pressure stage and comprises eight compressor modules **100A-H** in one or two stacks **201D** totaling sixteen compressor heads **31, 51**, with each compressor head configured to pressurize process gas to 15,000 psi. Generally, the stages **202A-D** may be referred to by their relative output pressure. One or both of the first and second stages **202A-B** may be considered “low pressure stages” while one or both of the third and fourth stages **202C-D** may be considered “high pressure stages;” alternatively, the second and third stages **202B-C** may be considered “medium pressure stages.” In some embodiments, the high-throughput compressor system **200** may increase the overall process gas throughput by including plumbing for a given compressor head **31, 51** to be used for multiple stages, providing selective gas configurability. Although the lower pressure heads **31, 51** may not be capable of use as high pressure stages when the gas discharge pressure is high, the higher pressure heads **31, 51** and components (e.g., high-pressure circuit **134** and/or medium-pressure circuit **132**) can be used as a lower pressure stages when the process gas discharge pressure of the system is low, and the overall pressure increase does not yet require as many compression stages. The reconfiguration of the gas compression heads to serve as different stages may provide an increase in flow throughput and consequently e.g. reduce tank filling times.

(109) In some embodiments such as FIG. **23**, the high-throughput compressor system **200** comprises four check valves **258** to implement the gas configurability option. This configuration allows the second stage **202B** to also selectively function as the first stage (e.g., outputting process gas at 1,000 psi), the third stage **202C** to also selectively function as the second stage (e.g., outputting process gas at 2,000 psi), and the fourth stage **202D** to also selectively function as the third stage (e.g., outputting process gas at 7,500 psi). This embodiment may result in a gain of up to about 15% in flow throughput over the course of a tank fill compared to sequentially running the stages **202A-D** individually (e.g. 11.5 kg/min vs 10 kg/min). When the outlet pressure is low (e.g. 30 to 85 bar for hydrogen process gas), only a single compression stage is needed, so all of the compressor heads **31, 51** may be plumbed in parallel and configured to provide the pressure of the first stage **202A**. When the outlet pressure increases (e.g. 85 to 250 bar), only two compression stages are needed, and roughly $\frac{3}{4}$ of the total compressor displacement may be used as the first stage **202A**, and the remaining $\frac{1}{4}$ may be used as the second stage **202B**. The increased flow rates of this embodiment are most significant when the system can operate as 1 or 2 stages, but this may only be possible for about 5% and 20% (respectively) of the total duration of the tank fill. Generally, the compressor heads **31, 51** of any stage **202A-D** are able to provide the selective operation at other desired pressures as long as sufficient pressure is applied to the diaphragm **5**; in certain embodiments this pressure is due to the suction condition applied to the compressor head, for example the suction pressure defined by the check valve **258** at the outlet port **7** (see also FIGS. **6, 15**)

(110) It will be appreciated that embodiments of the present disclosure may comprise various numbers and physical arrangements of stages **202**, stacks **201** per stage **202**, compressor modules **100** per stack **201**, compressor modules **100** per stage **202**, compressor heads **31, 51** per compressor module **100**, and compressor heads **31, 51** per stage **202**. Moreover, embodiments may have various pressure ratings of the compressor heads **31, 51**. Accordingly, embodiments of the diaphragm compressor system **200** comprise from one to ten stages **202** with particular embodiments comprising one, two, three, four, five, six, seven, eight, nine, ten or more stages **202**. Embodiments of the diaphragm compressor system **200** comprise from one to ten stacks **201** or more and any number of the stacks may comprise one or more stages **202** (for example, stack **201**

in FIG. 1 may be configured with compressor modules **100A-B** comprising a first stage **202A** and compressor modules **100C-D** comprising a second stage **202B**). In some embodiments, an individual stack **201** comprises one, two, three, or more stages **202**. Embodiments of the diaphragm compressor system **200** comprise from one to twelve or more compressor modules **100** per stack **201** and, similarly, one to twenty-four compressor heads **31, 51** per stack, or any ranges therebetween. Other embodiments comprise stacks **201** with different numbers of compressors **100** or compressor heads **31, 51** per stack. Certain embodiments of the diaphragm compressor system **200** comprise from one to six compressor heads **31, 51** per compressor module **100**.

(111) The output performance of embodiments of the diaphragm compressor system **200** may likewise have various configurations system-wide and various output configurations among the constituent compressor heads **31, 51**, modules **100**, and stages **202**. For compressed hydrogen process gas, embodiments of the diaphragm compressor are configured to output pressures up to 30,000 psi or more. Embodiments of the diaphragm compressor system **200** comprise stages **202** that each have a compression ratio in a range of about 1:1 to 10:1 or a range of about 2:1 to 6:1; such ratios may be distinct from each other. In certain embodiments, the compressor system **200** comprises a first stage **201A** outputting process gas at about 40-7,500 psi and an additional stage (e.g., second stage **201B**, third stage **201C**, and/or fourth stage **201D**) outputting process gas at about 1,000-15,000 psi. In other embodiments, the compressor system **200** comprises a first stage **201A** outputting process gas at about 100-7,500 psi, optionally a second stage **201B** outputting process gas at about 200-15,000 psi, optionally a third stage **201C** outputting process gas at about 300-25,000 psi, and optionally a fourth stage **201D** outputting process gas at about 400-30,000 psi.

(112) In embodiments, the high-throughput compressor system **200** is configured for a tank-filling operation from 30 to 1000 bar, with a 4-stage system comprising stages **202A-D**. This is generally similar to FIG. 23, comprising a first stack **201A** of compressor modules **100** comprising a first stage **202A** of the lowest pressure, a second stack **201B** of compressor modules **100** comprising a second stage **202B** of a higher pressure than the first stack, a third stack **201C** of compressor modules **100** comprising a third stage **202C** of a higher pressure than the second stack, and a fourth stack **201D** of compressor modules **100** comprising a fourth stage **202D** of a highest pressure.

(113) Disclosed embodiments and features of the high throughput compressor system **200** for hydrogen process gas can meet a throughput target of up to 10 kg/min or more compressed hydrogen at a minimum outlet pressure of 875 bar, with embodiments capable of 1,000 bar or more. Embodiments provide a compact compressor module **100** that can be stacked together with one or more additional compressor modules and plumbed to achieve essentially any required throughput. For some embodiments, the design is a system with two stages **202A-B** with approximately sixteen diaphragms **5** (each compressor head **31, 51** having one diaphragm **5**) per stage **202**, which may be eight compressor modules **100** per stage and is designed for 120 bar inlet pressure. The present disclosure can be applied to a larger system with lower inlet pressure such as 30-50 bar inlet pressures, requiring four stages **202A-D** to achieve compression up to 1,000 bar.

(114) Certain embodiments provide two stacks **201** of four compressor modules **100** per each stage **202**, although other arrangements are feasible and contemplated. Each module **100** may actuate two compressor heads **31, 51** of the same size and operating pressures, as would all modules **100** within the same stack **202**, and thus have common suction and discharge gas pressures. This has benefits for simplicity of gas plumbing (e.g., hydraulic components such as HPU **118**, pressure circuits **130, 132, 134, 138** or main stage valve **250** are the same and may be operatively connected to multiple compressor modules **100**), and also reduction in accumulator count and size (e.g., one or more of accumulators **136A-E** (see, e.g., FIGS. 2-3, 15).

(115) In some embodiments, one of the compressor modules **100** can be deactivated within the stack **201** and still allow the stack to operate. The compressor module may be deactivated for servicing, repair, or replacement, e.g. by isolating the module **100** by valving. Because the rest of the modules **100** in the stack **201** continue to operate, maintenance can be temporarily deferred for

a more convenient time if desired. Additionally, for some embodiments such as embodiments with multiple stacks **201** per stage **202**, the compressor system **200** may provide continued operation during service, albeit at a reduced throughput. This could be achieved by valving and deactivating one stack **201A** while the other stack **201B** of the stage **202A** continues. Consequently, the other stacks **201A** of other stages **202B** before or after the compressor module being serviced may need to be shut down accordingly to match pressure ratios per stage, but this may nonetheless allow continued operation during service and make emergency service requirements less detrimental to overall system operation. Effectively, by having multiple stacks **201** per stage **202**, the system may create a redundancy effect which is beneficial from a failure and service perspective.

(116) Damping of the Hydraulic Actuator

(117) As shown generally in FIGS. **6-11** with certain embodiments detailed in FIGS. **8** and **11**, certain embodiments of the compressor module **100** comprises a damping mechanism **105** that includes venting of the work oil being compressed in the direction of travel of the actuator piston **126**. Generally, in some embodiments, the actuator piston travel distance in a discharge stroke may be about 0.5-3 inches, about 1-2.5 in., about 0.5 in., about 1 in., about 1.5 in., about 2 in., about 2.5 in., about 3 inches, or about 0.5 to 4 inches. In embodiments, the travel time of the actuator piston is less than 100 milliseconds (ms). In certain embodiments, the travel time is about 30-95 ms, about 45-75 ms, about 50-70 ms, or about 60-65 ms. Additionally, in embodiments the dwell time of the actuator piston **126** is less than about 50 ms, less than about 25 ms, about 5-30 ms, about 10-25 ms, or about 15-20 ms. Accordingly, the actuator piston **126** reciprocates with quick starts and quick stops including possible impact with a hard stop **106** formed in the drive housing **114**.

Embodiments of the present disclosure comprise a damping mechanism **105** to aid in stopping the actuator piston **126** as it approaches the end of a stroke to decrease the impact velocity against the hard stop **106**. The damping mechanism **105** is provided on both sides of the actuator piston **126** and in each of the first and second actuation volume **144**, **146**.

(118) The drive housing **114** comprises a plurality of ports **147** including the first and second distal ports **148A**, **148B** that are in fluid communication with components of the hydraulic drive **110**. The hydraulic drive **110** and the HPU **118** are configured to provide a variable-pressure supply of work oil to the drive cavity **116** through one or more of the plurality of ports **147**. One or more of the plurality of ports **147** is configured to supply work oil to the hydraulic drive **110**, and one or more of the plurality of ports is configured to vent work oil out of the hydraulic drive. In the illustrated embodiments, the plurality of ports **147** is configured to both supply and vent work oil from the hydraulic drive **110** depending on the direction of travel of the actuator piston **126**, as with the first and second distal ports **148A**, **148B** of FIGS. **6-8**. The first and second distal ports **148A**, **148B** are each operatively coupled to a respective main stage valve **250** to control the supply and vent operations.

(119) Referring to FIGS. **6-8**, the drive housing **114** comprises orifices **152** for both the first and second actuation volumes **144**, **146**. The orifices **152** are operatively connected to either a first radial port **153** at the first actuation volume **144** or a second radial port **155** at the second actuation volume **146**. In other embodiments and as shown in FIGS. **10-11**, the damping mechanism **105** comprises a first plurality of orifices **152A** and a second plurality of orifices **152B** in communication with the drive cavity **116**. In embodiments, the orifices **152**, **152A**, **152B** are also in communication with one or more ports of the plurality of ports **147**. In embodiments, one or more of the orifices **152**, **152A**, **152B** are in communication with the plurality of ports **147** for both venting and supply of work oil. In certain embodiments, the orifices **152**, **152A**, **152B** at the first actuation volume **144** are in fluid communication with the first distal port **148A** and the orifices **152**, **152A**, **152B** at the second actuation volume **146** are in fluid communication with the second distal port **148B**. For any such embodiments, the work oil that is vented from the drive cavity **116** through the orifices **152**, **152A**, **152B** and one or more of the plurality of ports **147** is supplied to the reservoir **230** or an accumulator such as the recovered oil accumulator **136D**.

(120) During the discharge cycle of the first compressor head **31**, the hydraulic drive **110** is configured to provide the variable-pressure supply of work oil through the second distal port **148B** to the second actuation volume **146** to press against the second side **145** of the actuator piston to drive the actuator piston, driving the first diaphragm piston **3** toward the corresponding first compressor head **31**, intensifying the work oil in the first variable volume region **54** to an intensified pressure, and actuating the diaphragm **5** of the first compressor head to the second position. As the actuator piston **126** moves, the damping mechanism **105** comprises the drive cavity **116** being configured to dampen the drive motion of the actuator piston **126** due to a volume of work oil in the opposing first actuation volume **144** with outflow restricted (i.e., the first actuation volume is in the direction of travel of the actuator piston). In certain embodiments, the volume of work oil vents through the first plurality of orifices **152**, **152A** and out of the first actuation volume **144**, providing space for the actuator piston **126**. Therefore, the damping force of the damping mechanism **105** is a function of the number and size of the plurality of orifices **152**, **152A**, (and **152B** discussed below), provided that the orifices freely flow to vent.

(121) At the beginning of the actuator piston **126** stroke for the discharge cycle of the first compressor head **31**, the first plurality of orifices **152**, **152A** is open to the first actuation volume **144**. Subsequently, the first plurality of orifices **152**, **152A** is progressively covered by the actuator piston **126** as it moves along its driving stroke, which constricts outflow through the first plurality of orifices and increases the damping force of work oil remaining in the first actuation volume **144** against the first side **143** of the actuator piston. In other words, as the obstruction by the actuator piston **126** occurs the effective size of the plurality of orifices **152**, **152A** decreases and the damping force increases because there is less area for the work oil in the first actuation volume **144** to escape. In this manner, the damping mechanism **105** provides an increasing damping force configuration due to work oil that remains in the first actuation volume **144** having access to less available venting area.

(122) In some embodiments and as shown in FIGS. **10-11**, the drive housing **114** further comprises a second layer of orifices illustrated as a plurality of second or supplemental orifices **152B** in communication with the first actuation volume **144**, the plurality of supplemental orifices being staggered axially relative to the plurality of first orifices **152A**. In the illustrated embodiments, the plurality of supplemental orifices **152B** are located relatively closer to the respective compressor head **31**, **51** and further along the discharge stroke of the actuator piston. In other embodiments, the plurality of supplementary orifices **152B** may partially overlap axially with the plurality of first orifices **152A**. The plurality of supplemental orifices **152B** dampen the driving of the actuator piston **126** due to the volume of work oil in the first actuation volume **144** that slowly vents through the plurality of supplemental orifices **152B** during driving of the actuator piston. As the actuator piston continues its stroke past the first plurality of orifices **152A**, the actuator piston **126** progressively obstructs the plurality of supplementary orifices **152B**. As the obstruction increases, the damping force increases due to work oil that remains in the first actuation volume **144** having access to less available venting area.

(123) In some embodiments, the second orifices **152B** are smaller than the first orifices **152A**, which smaller diameter provides a relatively higher damping force due to less available venting area compared to an equal number of first orifices. When arranged as shown in FIG. **11**, as the actuator piston **126** completes its stroke from right to left, the damping mechanism **105** provides an increasing damping force configuration due to several factors: progressively obstructing the first plurality of orifices **152A**, completely blocking the first plurality of orifices **152A**, the remaining second plurality of orifices **152B** being relatively smaller, progressively obstructing the second plurality of orifices **152B**, and finally completely blocking the second plurality of orifices **152B**. Accordingly, the damping mechanism **105** increases the damping force against the actuator piston **126** as it nears the end of its stroke.

(124) Embodiments of the first and second orifices **152A-B** and their associated porting (including

the plurality of ports **147**) may have various shapes, sizes, and orientations. In embodiments one or both of the first and second orifices **152A-B** are circular, though other embodiments may be elongated in the direction of actuator piston driving, for example oval shaped with a long axis parallel to the direction of actuator piston **126** driving. In some embodiments, the first and second orifices **152A-B** are formed in one or more surfaces of the drive housing **114** oriented at a non-parallel angle relative to the actuator piston axis **208**. In embodiments, the orifices **152A-B** are formed in surfaces extending substantially perpendicular to the actuator piston axis **208**. In some embodiments, the first and supplemental orifices **152A-B** extend radially away from the drive cavity **116** and the actuator piston **126**.

(125) In certain embodiments, 24 orifices **152** are provided in the drive cavity **116**, with 12 orifices in the first actuation volume **144** and 12 orifices in the second actuation volume **146**. Similarly, in embodiments, up to 24 first orifices **152A** and 24 second orifices **152B** are formed in the drive cavity **116** or more generally, in embodiments the number of orifices **152** may be any number from 1-48 orifices. In other embodiments, the number of orifices **152** may be greater or smaller, such as each actuation volume **144**, **146** having up to 100 or up to 200 orifices or more. In embodiments, additional layers of orifices **152** may be included. The number of orifices may be different in different layers, for example the number of first orifices **152A** may be different than the number of second orifices **152B**.

(126) Referring to FIGS. 7-8, in embodiments the drive housing **114** comprises a slight annular gap **151** between the actuator piston **126** and the drive cavity **116** and extending around an outer surface of the actuator piston (e.g., the circumferential outer surface in the illustrated embodiment). The annular gap **151** is in fluid communication with both the actuation volume **144** and the second actuation volume **146** and, in some embodiments, is configured to dampen the driving of the actuator piston **126** throughout the piston stroke in either direction by maintaining a small volume of work oil that is not in direct communication with any of the plurality of ports **147**. Accordingly, in certain embodiments, the annular gap **151** is positioned to dampen the driving of the actuator piston **126** after the orifices **152** are obstructed by the actuator piston. When the orifices **152** are fully closed leaving the plurality of ports **147** unable to vent, the relatively small annular recess **151** provides a small amount of flow area and acts like a fixed orifice during final damping. As shown in FIG. 8, as the actuator piston **126** strokes to the left, work oil leaves the actuation volume **144** via the orifices **152**. In certain embodiments, the circumferentially-arranged orifices **152** are fully closed off at the end of stroke; in other embodiments the orifices are fully closed off slightly before. The dampening capability of the annular gap **151** is at least in part due to compressibility of the work oil.

(127) Referring to FIGS. 9-11, in embodiments, additional or final damping is provided by venting work oil through the first opening **154** and the internal porting **127A**, **127B** of the actuator piston **126**. As shown, the internal porting **127A** is in fluid communication with the plurality of ports **147**, in particular first proximal port **148C** and (indirectly) first distal port **148A**. In embodiments, the first and second proximal ports **148C**, **148D** are low-pressure ports supplying the low-pressure rail **130** and comprising a check valve preventing a vent flow out of the drive cavity. A landing orifice **107** connects the first proximal port **148C** to the first distal port **148A**, and vented work oil from the internal porting **127** flows out through the first distal port **148A** to a pressurized circuit, accumulator, or the reservoir **230**.

(128) The landing orifice **107** is configured (e.g., sized) to provide desired deceleration performance of the actuator piston **126** at the end of its stroke in landing against the hard stop **106** with requisite velocity for intensifying process gas without excessive velocity that may damage components or otherwise inhibit operation. In some embodiments, the first opening **154** and the internal porting **127** vent work oil from the accumulation volume **144** throughout the stroke of the actuator piston; this venting through the first opening may occur during and/or after venting through the first and second plurality of orifices **152A**, **152B**. In certain embodiments, the internal

porting **127** is configured to vent work oil after the first and second orifices **152A-B** have been completely blocked. In embodiments, the landing orifice **107** is configured to vent only after the first and second orifices **152A-B** by comprising a check valve (not shown) with a threshold pressure set at a relatively high pressure that is only achieved after the first and second orifices **152A-B** have been completely blocked. Similarly at the opposite end of the actuator piston **126**, in embodiments the internal porting **127B** is in fluid communication with the plurality of ports **147** and an additional landing orifice **107**.

(129) In some embodiments, aspects of the damping mechanism **105** may be customized or tuned to increase or decrease the damping force against the actuator piston **126**. In some embodiments, the landing orifice **107** and/or one or more of the orifices **152** may comprise a removable orifice (not shown) that can be exchanged for orifices of different size or flowrate. One or more orifices **152** may comprise a removable plug (not shown) to block one or more of the orifices by switching out. In embodiments, one or more of the annular recess **151** or the first and second plurality of orifices **152** are configured to be removable from the drive housing **114** individually or as a ring of orifices. In embodiments, this customization and tuning may optimize performance of the same compressor module **100** in different use cases (e.g., in different stack and staging arrangements or for different process gas output pressures) or in different environments (e.g., in different elevations or climates). In certain embodiments, this customization and tuning may optimize performance of the same drive housing **114** and drive cavity **116** for different pressure ratings of the compressor heads **31**, **51**. The first and second proximal ports **148C**, **148D** may be operatively connected to the low-pressure circuit **130** or the medium-pressure circuit **132**.

(130) In some embodiments, the drive housing **114** further comprises a removable sleeve insert **115** mounted internally to define the drive cavity **116** and may comprise the plurality of orifices **152**, **152A**, **152B** and the first and second radial ports **153**, **155** in whole or in part, along with other components of the drive housing **114**. The sleeve insert **115** is subjected to significant loads and wear forces due to the motion of the actuator piston **126** and the pressurization of the drive cavity **116**. Therefore, the sleeve insert **115** is removable for replacement after wearing down without requiring replacement of the whole drive housing **114**. In embodiments and as illustrated in FIGS. **6-11**, the sleeve insert **115** comprises one or more annuli **149A-D** in fluid communication with the drive cavity **116** and a respective one or more of the plurality of ports **147**. In certain embodiments, a first and second distal annulus **149A**, **149B** are arranged respectively with the first and second distal ports **148A**, **148B** and, similarly, a first and second proximal annulus **149C**, **149D** are arranged respectively with the first and second proximal ports **148C**, **148D**. In other embodiments not illustrated, one or more of the plurality of ports **147** does not have a corresponding annulus and instead extends through the drive housing **114** and the sleeve insert **115** directly to the first or second actuation volume **144**, **146**.

(131) Each of the annuli **149A-D** extends partially or completely around the actuator piston **126** and operatively couple together the multiple discrete ports that constitute the respective port and the multiple orifices that constitute the plurality of orifices **152**, **152A**, **152B**. For example in FIGS. **10-11**, the first distal annulus **149A** connects the multiple first distal ports **148A** with both of the first plurality of orifices **152A** (corresponding radial ports not shown) and the second plurality of orifices **152B** (through first radial ports **153**) and the first proximal annulus **149C** connects the multiple first proximal ports **148C** with the first internal porting **127A**. In the example of FIG. **7**, the first distal annulus **149A** connects each of the multiple first distal ports **148A** arranged around the actuator piston **126** with the manifold port **117**.

(132) In embodiments with first and second compressor heads **31**, **51**, during the discharge cycle of the second compressor head **51**, the drive cavity **116** is configured to similarly dampen the driving of the actuator piston **126**. A volume of work oil in the second actuation volume **146** provides damping force and vents through the plurality of orifices **152**, **152A** during driving of the actuator piston **126**. The plurality of orifices **152**, **152A** are open to the second actuation volume **146** when

the driving of the actuator piston **126** begins, and the plurality of orifices are progressively covered by the actuator piston during the driving. Covering the plurality of orifices increases the damping force of work oil remaining in the second actuation volume **146** acting against the second side **145** of the actuator piston **126**. In embodiments, the internal porting **127B** and corresponding landing orifice **107** provide damping and control the final velocity of the actuator piston **126** impacting the respective hard stop **106**.

(133) It will be appreciated that in embodiments, any of the ports and orifices that provide damping can be reversible and configured to provide an actuation supply of pressurized work oil for the actuator piston **126**, for example in a discharge cycle of the second compressor head **51**. In embodiments, the actuation supply is provided sequentially in a reverse order from the damping sequence above. Accordingly, for the discharge cycle of the second compressor head **51** in FIGS. **9-11**, the actuation supply of work oil to the first actuation volume **144**, work oil (e.g., low-pressure work oil from the low-pressure circuit **130**) begins from the first proximal port **148C** and then through the first internal porting **127A** of the actuator piston **126** feeding to the first opening **154**. Subsequently, the second plurality of orifices **152B** is configured to provide additional actuation supply of pressurized work oil via the first distal port **148A**. Finally, the first plurality of orifices **152A** is configured to provide additional actuation supply of pressurized work oil via the first distal port **148A**. In embodiments, this sequential actuation supply provides one or more of the plurality of pressure circuits **120**. In some embodiments, one or more ports of the plurality of ports **147** is configured to further supplement the actuation supply of work oil provided by another port. In certain embodiments the first proximal port **148C** comprises a bypass check valve (not shown) to provide supplemental flow in addition to the first distal port **148A**, which may be configured to avoid cavitation of work oil in the first actuation volume **144** as the actuator piston **126** moves quickly. As discussed above, the initial movement of the actuator piston **126** may be aided by additional forces such as the return of the diaphragm **5** of the first compressor head **31**.

(134) The damping mechanism **105** may be advantageous for embodiments of the compressor module **100** with a short stroke of the actuator piston **126**, for example about 1" or 2.5", or more generally a stroke below about 7.5". With shorter travel distance, the peak speed of the actuator piston **126** may be lower than with a relatively longer stroke, and deceleration at the end of the stroke may achieve low impact velocities. In certain embodiments, compressor heads **31**, **51** that are configured for higher pressures comprise a stroke of about 1" (e.g., about 7,500 psi and above, including embodiments comprising about 7,500 psi and about 15,000 psi and ranges therebetween), whereas compressor heads configured for relatively lower pressure comprise a stroke of about 2.5" (e.g., about 5,000 psi and below, specific embodiments comprising 1,000 psi; 2,000 psi; 5,000 psi; and ranges therebetween).

(135) As discussed above, in certain embodiments of the damping mechanism **105**, the actuator piston **126** closes off circumferential primary supply ports (such as the first and second distal ports **148A**, **148B**) as it reaches the hard stop **106**. Dynamic computer simulations of such embodiments of the damping mechanism **105** show that, for embodiments of the actuator pistons **126** with 1" stroke, the impact velocities are less than about 0.2 m/s for nominal conditions and less than about 0.7 m/s for avoiding a failure condition. The simulations assumed nominal radial clearances in a range between about 0.0025-0.010 inches between the actuator piston **126** and inner walls of the drive cavity **116**, although smaller clearances are contemplated for other embodiments. For the actuator pistons **126** with 2.5" stroke and higher pressure supply, the impact velocities were less than about 0.3 m/s for nominal conditions and less than about 1.5 m/s for avoiding a failure condition. It will be appreciated that other embodiments may comprise a broader range of values for parameters such as the stroke distance of the actuator piston **126**, landing velocity, and output pressure of the compressor heads **31**, **51**.

(136) Main Stage Valve

(137) In certain embodiments, the high-throughput compressor system **200** comprises one or more

main stage valves **250** ("MSV **250**") to control the hydraulic drive **110**, in particular the flow of work oil to and from the drive cavity **116**.

(138) The work oil drives and damps the actuator piston **126**, therefore the MSV(s) **250** control the timing (cycle time, travel time) of the actuator piston **126**. As detailed above, the actuator piston **126** discharge stroke travel distance may be in a range of about 0.5-3 inches with a travel time of less than 100 milliseconds, other embodiments may range from 0.5-7 inches or more. Accordingly, the timing of the MSV(s) **250** in supplying and venting work oil from one or more of the plurality of pressure circuits **120** must correspond to these parameters.

(139) In some embodiments, the MSV(s) **250** control the interface of the HPU **118** and of the one or more pressure circuits **120** with the hydraulic actuator **112**, such interface including both the supply and vent of work oil for the drive cavity **116**. In other words, the MSV(s) control a pressurized hydraulic supply of work oil for operating the hydraulic actuator **112** and the MSV(s) may control at least some venting of work oil from the drive cavity **116**. In embodiments, the MSV **250** is an actively-controlled valve. In the illustrated embodiment, the MSV **250** is a three-way valve as shown in FIGS. **18A** (vent stage) and **18B** (supply stage).

(140) Referring to FIGS. **18A-B**, in embodiments, The MSV **250** for controlling a diaphragm compressor system **100** comprises a valve body **260** comprising a first end **262** and a second end **264**, a pilot port **266** proximate the first end, a supply port **268**, a first vent port **270**, a second vent port **271**, and a cylinder port **272**. The MSV **250** comprises a pin subassembly **274** for reciprocating in the valve body **260**, the pin comprising a spool **276**, a pilot pin **278** proximate the second end **264**, and a return pin **280** proximate the first end **262**. In certain embodiments, the MSV **250** is mounted to the valve manifold **244** or the drive housing **114**. In some embodiments, each of the vent port **270** and the cylinder port **272** are in fluid communication with the drive cavity **116** and the supply port **268** operatively coupled to the hydraulic power unit **118**. In embodiments, the position of the pin subassembly **274** selectively blocks one or both of the vent port **270** and the cylinder port **272** to control the flow of work oil between the drive cavity **116** and the MSV **250** along with controlling the flow of work oil between the MSV **250** and any other component(s) attached to the MSV. It will be appreciated that any of the ports may be one or more ports in some embodiments, and in the illustrated embodiment each of the pilot port **266**, supply port **268**, the first vent port **270**, and the cylinder port **272** is a plurality of ports arranged annularly about the pin subassembly **274**.

(141) FIG. **18A** shows a vent position of the MSV **250** configured for allowing an outflow of work oil from the drive cavity **116**, e.g. from work oil in the first or second actuation volume **144**, **146** when being compressed by the actuator piston **126** and vented through the plurality of orifices **152**. The pin subassembly **274** is configured to move axially to the vent position with the cylinder port **272** in fluid communication with the first vent port **270**, such that work oil from the drive cavity **116** flows to the cylinder port **272**, through the valve body **260**, and out through the first vent port **270**. In embodiments, the hydraulic drive **110** further comprises a recovered oil accumulator **136D** operatively coupled to the first vent port **270** of the MSV **250** for storing and recycling this work oil in subsequent cycles. In other embodiments, the first vent port **270** is operatively connected to a reservoir **230** of the hydraulic drive **110**.

(142) In embodiments, the MSV **250** comprises one or more vent orifices **282A**, **282B** configured to vent work oil out of the MSV and dampen motion of the pin subassembly **274** when moving into the vent position. The vent orifices **282A**, **282B** are arranged annularly around the pin subassembly **274**. Similar to the damping mechanism **105** of the actuator piston **126**, the vent orifices **282A** in the MSV **250** are configured to be progressively obstructed as the pin subassembly **274** moves axially, increasing the damping force as the pin reaches the end of its stroke. Accordingly, the pin subassembly **274** moves quickly to the vent position without any hard impact or bounce. In the illustrated embodiment, vent orifices **282B** are not configured to be obstructed, but in other embodiments these or additional layers of orifices (not shown) may be configured to be obstructed

in addition to the vent orifices **282A**.

(143) FIG. **18B** shows a supply position of the MSV **250** configured to supply work oil to the hydraulic actuator **112**. The MSV **250** is configured to selectively move to the supply position to operatively connect the HPU **118** to the drive cavity **116** during the discharge cycle of the first or second compressor head **31**, **51**. In embodiments, the MSV **250** supply position is configured to control a discharge stroke of the compressor head **31**, wherein the MSV **250** connects one of the pressure circuits **120** to the second actuation volume **146** of the drive cavity **116** to supply pressurized work oil to the second side **145** of the actuator piston **126** to drive the actuator piston and consequently drive the diaphragm piston **3** toward the diaphragm **5** of the first compressor head **31**. This connection in embodiments is from the high-pressure circuit **134**, medium-pressure circuit **132**, or low-pressure circuit **130**. The pin subassembly **274** is configured to move axially to the supply position with the supply port **268** in fluid communication with the cylinder port **272**, such that work oil flows from the HPU **118** or a pressure circuit enters through the supply port **268**, passes through the valve body **260**, and exits through the cylinder port **272**. An end orifice **284** is located proximate the first end **262** of the MSV **250**. The pilot port **266** and the end orifice **284** configured to vent work oil out of the MSV **250** and dampen motion of the pin subassembly **274** when moving into the supply position in a similar manner to the vent orifices **282A**, **282B** for the vent position.

(144) In embodiments, during a suction cycle of the first compressor head **31**, the MSV **250** is configured to move to the vent position (FIG. **18A**) to connect the drive cavity **116** of the drive housing **114** to the first vent port **270** of the main stage valve **250**, and the hydraulic drive **110** vents work oil from the second actuation volume **146** to the main stage valve **250** through the vent port **270**.

(145) In embodiments, one or more of the pilot port **266** and the vent orifice **282A-B** comprises a plurality of rows of orifices that are axially spaced. In certain embodiments, this plurality of rows comprises a row of relatively larger orifices proximate the spool **276** and a row of smaller orifices proximate the respective first or second end **262**, **264**. In some embodiments, the pilot port **266** comprises a ring or layer of removable orifices, or plugs for certain orifices, for fine tuning of the damping performance. Such tuning may be necessary for implementing the MSV **250** with different pressure ratings of compressor heads **31**, **51**, for utilizing different types of work oil, or for operating at different temperature ranges.

(146) In embodiments, the low-pressure circuit **130** of the hydraulic drive **110** further comprises a recovered oil accumulator **136D** operatively coupled to the first vent port **270** of the main stage valve **250**. In such embodiments, in the vent position, the main stage valve **250** is configured to supply oil from the drive cavity **116** to the cylinder port **272**, through the valve body **260**, and exiting the first vent port **270** to the recovered oil accumulator. In some embodiments, the low-pressure circuit **130** comprises the recovered oil accumulator **136D**. In certain embodiments, a passive valve **131** (FIG. **17**) is operatively connected to the recovered oil accumulator **136D** and the drive cavity **116** downstream of the main stage valve **250**. During the suction cycle of the first compressor head, the passive valve **131** is configured to supply oil from the recovered oil accumulator **136D** to the drive cavity **116**. In embodiments, this supply from the recovered oil accumulator **136D** may occur at one or more times, for example during a low-pressure stage of tank fill or during a beginning portion of each actuator **126** stroke.

(147) In some embodiments, the MSV **250** further comprises a pilot valve **290** configured to selectively actuate the pin subassembly **274** of the MSV **250**. The pilot valve **290** controls a supply of pilot fluid (e.g., work oil at a pilot pressure) to the MSV **250** to move the pin subassembly **274**. The pilot valve **290** in the illustrated embodiment is mounted in the second end **264** of the valve body **260** and is a multi-stage valve comprising a spool and two coils. In some embodiments, the hydraulic drive **110** further comprises a pilot pressure circuit **138** and a pilot pressure accumulator **136E** operatively coupled to the pilot valve. The pilot pressure accumulator **136E** in embodiments

is charged in various ways such as a separate hydraulic unit, the HPU **118**, or recovered intensified work oil vented from one or more MSVs **250**. In some embodiments, the pilot valve **290** is a three-way valve or two two-way valves. In other embodiments, a different actuator or valve (not shown, for example a spool valve with one coil and one spring return, a piezoactuator, or servomotor) is implemented with the MSV **250** to selectively actuate the pin subassembly **274** of the MSV **250** to the supply position

(148) In embodiments, the pilot pressure circuit **138** is also operatively coupled to the pilot port **266** at the first end **262** of the valve body **260**. In some embodiments, the pin subassembly **274** of the MSV **250** has a larger area proximate the pilot valve **290** than proximate the pilot port **266**. Therefore, when pilot pressure is supplied to the pilot pin **278** through the pilot valve **290** and the pilot port **266**, the pin subassembly **274** is configured to move to the supply position. In embodiments, the MSV **250** comprises a return spring **286** configured to bias the pin subassembly **274** toward the vent position when pressure is not supplied to the pilot valve **290**. As shown in FIG. **15**, each of multiple MSVs **250A-D** may comprise a respective pilot valve **290**.

(149) As noted above, some embodiments of the high-throughput compressor system **200** and/or the individual compressor modules **100** comprise multiple MSVs **250**. Referring also to FIGS. **15-17**, in embodiments a first MSV **250A** is operatively coupled to the first actuation volume **144** of the drive cavity **116**. A second MSV **250B** is substantially similar to the first MSV **250A**, and the second MSV **250B** is mounted to the drive housing **114** with each of the vent port **270** and the cylinder port **272** in fluid communication with the second actuation volume **146** of the drive cavity **116** and the supply port **268** operatively coupled to the hydraulic power unit **118**. The first MSV **250A** is configured to selectively move to the supply position to connect the high-pressure circuit **134** to the second actuation volume **146**, while the second MSV **250B** is configured to selectively move to the respective supply position to connect the medium-pressure circuit **134** to the second actuation volume **146**.

(150) In other embodiments, four MSVs **250A-D** are provided with a first MSV **250A** connecting the high-pressure circuit **134** to the first actuation volume **144**, a second MSV **250B** connecting the high-pressure circuit **134** to the second actuation volume **146**, a third MSV **250C** connecting the medium-pressure circuit **132** to the first actuation volume **144**, and a fourth MSV **250D** connecting the medium-pressure circuit **132** to the second actuation volume **146**.

(151) In embodiments one or more of the MSVs **250A-D** vents to the recovered oil accumulator **136D**. In the illustrated embodiment, the third and fourth MSVs **250C**, **250D** are each configured to vent from the respective first or second actuation volume **144**, **146** through the respective first vent port **170** to the accumulator **136D**.

(152) Referring to FIG. **20**, the valve manifold **244** is illustrated with some of the corresponding hydraulic components and internal plumbing. The MSV **250** is mounted in a valve mount **292** that is plumbed to the operative ports of the MSV **250**. The low-pressure circuit **130** is collected from several sources including each MSV **250** along with vented return from the hydraulic drive **110**, these sources lead to the recovered oil accumulator **136D**. The high-pressure circuit **134** is ported externally from the HPU **118** (not shown, see also FIG. **29**). Embodiments comprising the medium-pressure circuit **132** are similarly supplied by another HPU **118**. It will be appreciated that the valve manifold **244** in embodiments may be operatively connected to multiple compressor modules **100**, reducing the overall footprint of the diaphragm compressor system **200**. In certain such embodiments, each MSV **250** is configured and operatively connected to the multiple compressor modules **100**. In other such embodiments, the valve manifold **244** comprises one or more additional MSVs for the additional compressor module(s).

(153) In some embodiments, the location and orientation of ports in the MSV **250** are selected to fit in the valve housing **244** (FIG. **20**) while accommodating nearby components. In certain embodiments, pilot port **266**, the supply port **268**, and/or the first and second vent ports **270**, **271** are arranged to allow work oil to enter and exit on a same side of the valve mount **292** (FIG. **20**)

that is opposite from the pilot pressure circuit **138** and other control components. Embodiments of the present disclosure provide sufficiently large flow areas through the ports to result in minimal or substantially no pressure drop of pressurized fluid passing through the MSV **250**. In some embodiments, the MSV **250** is configured to accommodate pressures up to 5,000 psi and provides an effective flow area (CdA, i.e., discharge coefficient×area) of about 300 mm.^{sup.2}. In embodiments, the MSV **250** provides a CdA of about 275-325 mm.^{sup.2}, about 250-350 mm.^{sup.2}, about 200-400 mm.^{sup.2}, about 100-500 mm.^{sup.2}, at least 200 mm.^{sup.2}, at least 250 mm.^{sup.2}, or at least 300 mm.^{sup.2}. In embodiments, the MSV **250** is configured to accommodate pressures up to 15,000 psi.

(154) In other embodiments, other valve types are employed in addition to or in lieu of the MSV **250**, including poppet, spool, directional, proportional and servo valves, among others. Different types of valves could be used as MSV **250** to operate the system differently. In some embodiments, proportional valves control the flow into the system with a fixed supply pressure. In this way the valve could be used to speed up or slow down the travel of the hydraulic drive actuator to fit a desired profile or to reduce the velocity of the actuator **112** as it nears top dead center or bottom dead center.

(155) In other embodiments, digital or on/off valves allow full flow to be supplied to (or vented from) the MSV **250** with a fixed flow area. As these valves open to the pressurized supply of work oil, the maximum flow area is exposed and allows full flow into the MSV **250** as dictated by the differential pressure across the valve. These valves are closed to shut off flow to the hydraulic actuator **112** for embodiments as a two-way valve. These valves can also vent the hydraulic actuator **112** for embodiments as a three-way valve. In still other embodiments, a variation of the digital on/off valve has multiple outlet ports that could be opened in series to allow flow to variable areas within the hydraulic drive. In this valve, the internal spool moves only a portion of its travel distance to open up flow to a single outlet port, then as the spool continues its travel additional outlet ports are opened. Operation of the digital valves can be achieved in several ways. In embodiments, the digital MSVs **250** are operated with a solenoid to drive the valve. In other embodiments, the digital MSVs **250** are operated with a set of two-way pilot valves to control the supply of pilot fluid to drive the valve spool. In other embodiments, the digital MSVs **250** are operated with a single three-way pilot valve to control the supply of pilot fluid to drive the valve spool. It will be appreciated that in embodiments, the MSVs **250** can be combinations of one or more of the above valve types.

(156) Active Oil Injection System

(157) In some embodiments, the diaphragm compressor **1** employs a hydraulic injection pump system **10**. The hydraulic injection pump system **10** comprises a pump **12**, at least one oil check valve **45** and a fixed setting oil relief valve **14** as illustrated in FIG. **33**. Other embodiments discussed below replace the fixed setting oil relief valve with a variable pressure relief valve **52** (“VPRV **52**”). The injection pump system **10** primary function is to maintain the required oil volume between the high-pressure oil piston **3** and diaphragm set **5**. During the compressor **1** (e.g., compressor head **31** or **51**) suction stroke, a fixed volume of work oil is injected into the work oil region **35** of the compressor **1**. This ensures a sufficient volume of oil is injected during each suction stroke to ensure the oil volume is maintained for proper compressor **1** performance.

(158) In certain embodiments the oil volume between the diaphragm piston **3** and diaphragm **5** is impacted by two modes of oil loss. The first mode of oil loss is annular leakage past the diaphragm piston **3** (also referred to as a high-pressure oil piston) back to the drive housing **114** or an oil reservoir. This annular leakage may be most significant on high pressure compressors **1** operating above 5,000 psi. In some embodiments, the annular leakage varies during operation of the diaphragm compressor **1**.

(159) The second mode of oil loss is defined as “overpump” which is hydraulic flow over the oil relief valve **14** that occurs every cycle during normal compressor **1** operation. The injector pump

system **10** is designed and operated to maintain an “overpump” condition through the relief valve **14** ensuring the diaphragms **5** are sweeping the entire compressor cavity **15** (i.e., completely or substantially discharging process gas from the process gas region **36**) thereby maximizing volumetric efficiency of the compressor **1**. Embodiments of the present disclosure comprise an injection pump system **10** that is actively controlled, referred to as an active oil injection system (“AOIS”) **30** as further discussed below.

(160) Some embodiments of the injection systems **10** are mechanically adjustable by a user to vary the injector pump's **12** volumetric flow rate into the compressor **1**. However, this requires manual observations and adjustment. An incorrect volumetric displacement from the injection pump system **10** that does not sufficiently account for oil losses can lead to various machine failures.

(161) In certain embodiments, the hydraulic relief valve **14** has a manually adjustable relief setting. These oil relief valves are set to a fixed oil relief pressure setting that is higher than the maximum process gas pressure. The maximum process gas pressure is the maximum expected pressure of the process gas for any particular use case. This elevated relief setting allows the diaphragm **5** to contact the process gas head support plate **6** firmly before any work oil flows over the relief valve **14**, thus, assuring a complete sweep of the entire volume of the head cavity **15** at the highest expected pressure of the process gas. When the diaphragm reaches the top of the head cavity **15**, the diaphragm piston **3** still has a pressure below the setting of the relief valve **14**. During this period, the work oil in the work oil region **35** compresses further and the hydraulic pressure rises above the compressor gas discharge pressure until it reaches the setting of the oil relief valve **14**. At this point, the relief valve **14** opens and oil, in the amount of the injection pump displacement (i.e., injection volume) less the annular leakage in the system, is displaced over the oil relief valve **14**. This oil flow out of the relief valve **14** is defined as overpump. Because the annular leakage may vary during operation, in some embodiments the injection volume does not correlate or loosely correlates to the volume of overpump flow through the relief valve **14**. In other embodiments, the injection volume corresponds or correlates to the volume overpump flow (for example, when the annular leakage has only minor variation, the annular leakage is variably estimated for different operating conditions, or the annular leakage is measured or otherwise detected).

(162) Certain embodiments of the present invention include an active oil injection system **30** (“AOIS”) in a diaphragm compressor **1**. The feedback and control of the AOIS **30** allow the compressor system **100** to minimize any excess energy used while ensuring the complete sweep of the diaphragm **5** discussed above.

(163) In certain embodiments, the compressor **1** forms a hydraulic circuit **50** connecting the outlet **34** of the work oil head support plate **8** to the inlet **33** of the work oil head support plate **8**. In those embodiments, the hydraulic circuit may also include an oil reservoir **38** configured to collect overpumped work oil from the work oil region **35** via the outlet **34** of the work oil head support plate **8**. By forming a hydraulic circuit, oil is circulated from the oil reservoir **38**, through the inlet **33** and into the work oil region, and then out the outlet **34** and back into the oil reservoir **38**. In another sense, work oil that exits the outlet **34** and passes through the oil relief valve **14** constitutes the overpumped work oil from the compressor **1**.

(164) In other embodiments, the hydraulic circuit also includes an AOIS **30** including a hydraulic accumulator **39** configured to provide a supply of supplemental work oil to the inlet **33** of the work oil head support plate **8**. In certain embodiments, the hydraulic accumulator **39** may be a hydraulic volume or any style of hydraulic accumulator **39** such as a bladder, piston, or diaphragm gas over fluid style hydraulic accumulator **39**. In still further embodiments, the AOIS includes an AOIS pump **40** in communication with the hydraulic accumulator **39**, the AOIS pump **40** configured to produce a variable volumetric displacement of the supplemental work oil from the oil reservoir **38** to the hydraulic accumulator **39** or directly to the inlet **33**. As used herein, variable volumetric displacement means that the AOIS **30** can provide a variable volumetric flow (i.e. injection quantities of supplemental work oil via the pump **40**) and/or an independently variable speed (i.e.,

flow rate via the motor **41**), to the work oil region **35** depending on the particular process conditions of the compressor **1** (e.g., compressor head **31**). This allows for variable injection quantities during the compressor's **1** operation to maintain the compressor's **1** oil volume most efficiently within the compressor **1**, and particularly the work oil region **35**. In certain embodiments, the AOIS **30** includes the AOIS pump **40** operatively coupled to the hydraulic accumulator **39**, and a motor **41** configured to power the AOIS pump **40** independently from the hydraulic drive **110**. In other words, the speed and control of the motor **41** is completely independent from, and not mechanically linked to, the hydraulic drive **110** that powers the diaphragm piston **3**.

(165) In certain embodiments, the AOIS **30** utilizes the existing pressure dynamics within the compressor **1** to satisfy the hydraulic flow requirements into the compressor **1**, and particularly into the work oil region **35**. As the compressor **1** transitions through its suction and discharge cycles, the AOIS pump **40** charges and discharges the hydraulic accumulator **39**. During the compressor's **1** suction stroke, this lower pressure condition within the compressor **1**, including the work oil region **35**, creates a positive pressure differential between the hydraulic accumulator **39** and the oil within the compressor head **31**, and particularly in the work oil region **35**. During this suction condition, hydraulic flow goes through the oil inlet check valves **45** and through inlet **33** into the work oil region **35** satisfying the injection event. During this time, the pump **40** may be continuously pumping into the hydraulic accumulator. During this discharge stroke, the hydraulic pressure within work oil region **35** is greater than the pressure in the hydraulic accumulator **39** therefore there is no flow from the hydraulic accumulator **39** into the compressor. At least one check valve **45**, and in some embodiments at least two check valves **45**, prevent backflow from the work oil region **35** into the hydraulic accumulator **39** and beyond. During this condition, the hydraulic flow from the AOIS pump **40** pressurizes the hydraulic accumulator **39** in preparation for the next injection event.

(166) Further embodiments include a variable pressure relieve valve (VPRV) **52**, which includes a pressure relief mechanism **42** operatively coupled to the work oil region **35** of the diaphragm cavity **15**, the pressure relief mechanism **42** including a pressure relief valve **43** in communication with the outlet **34** of the work oil head support plate **8** and configured to relieve an outlet volume of the pressurized work oil from the work oil region **35**. In these embodiments, the pressure relief valve **43** includes a hydraulic relief setting corresponding to an overpump target condition of the pressurized work oil relative to the process gas discharge pressure. In some embodiments, the overpump target condition corresponds to a maximum process gas discharge pressure. In other words, the overpump target condition corresponds to a maximum process gas discharge pressure that the compressor head **31** is configured to operate at, so that the process gas region **36** is configured to be completely evacuated by the diaphragm **5** at maximum gas discharge pressure.

(167) In certain embodiments, during an oil relief event during the discharge cycle, the relief valve **43** opens and oil, in the amount of the injection volume per revolution less the annular leakage in the system, is displaced over the oil relief valve **14**, defined as overpump. During this time, the hydraulic flow from the AOIS pump **40** pressurizes the hydraulic accumulator **39** in preparation for the next injection event during the next suction cycle.

(168) However, in certain embodiments, the pressure relief valve **43** is configured to actively adjust the hydraulic relief setting of the pressure relief valve to correspond to an overpump current condition. In other words, the pressure relief valve **43** is configured to adjust the hydraulic relief setting up or down corresponding to a relative increase or decrease in gas discharge pressure. This prevents the compressor head **31** from experiencing more overpump than necessary to completely evacuate the process gas region **36** by the diaphragm **5** under conditions with a gas discharge pressure less than the maximum gas discharge pressure. Adjustability of the hydraulic relief setting may enable longer machine life expectancy and better system efficiency due to lower cyclic stresses and lower alternating loads during the compressor's **1** discharge and suction cycles.

(169) Certain embodiments of the AOIS **30** include an injector pump **40** and hydraulic accumulator

39 without a VPRV **52**, while other embodiments include both systems.

(170) In certain embodiments, the AOIS **30** includes a feedback mechanism configured to control the AOIS pump **40** to maintain the overpump target condition of the work oil region **35**. The feedback mechanism includes a measurement device **44** that provides feedback to verify the over pump condition is being met to control the injector pump system **30**. In certain embodiments, the feedback mechanism includes a first measurement device **44** operatively coupled to the diaphragm compressor **1**, the measurement device configured to detect and/or measure the overpump current condition of the intensified work oil flowing out of the outlet **34** from the work oil region **35**. In certain embodiments, the feedback mechanism is configured to adjust the volumetric displacement of the injector pump **40** to the hydraulic accumulator **39** in response to the overpump current condition.

(171) Turndown ratio refers to the operational range of a device, and is defined as the ratio of the maximum capacity to minimum capacity. In certain embodiments of the AOIS **30**, the AOIS is configured to provide a large turndown ratio of supplemental work oil relative to the work oil **4** in the work oil region **35** of the compressor **31**. By separating the functions of the hydraulic drive **31** and the AOIS pump **40**, a large turndown ratio can be achieved allowing for significant adjustability of injection quantity to tightly control the amount of overpump through the relief valve **43** over a wide range of operating conditions.

(172) In embodiments, the overpump target condition ranges from 0.1%-500% above a measured process gas discharge pressure. In various embodiments, the overpump target condition ranges from about 0.1%-100% above, 0.1%-50% above, 0.1%-40% above, 0.1%-30% above, 0.1%-20% above, 1%-20% above, or 1%-50% above the measured process gas discharge pressure.

(173) Alternative Embodiments

(174) Some embodiments incorporate commonality of parts and assemblies between stages **202** even though the later stages may have larger compressor heads, higher pressure rails/circuits, and the like. Specifically, such items as valve manifolds **244**, MSVs **250**, and other hydraulic components can be common for cost and simplicity purposes. Additionally, the clamping mechanism **204** may have duplicate components or similar primary components with minor deviations to accommodate adapting and mating to specific stages.

(175) In other embodiments, the first and second compressor heads **31**, **51** are driven by two separate hydraulic actuators **112** instead of a single hydraulic actuator, and the two hydraulic actuators may be configured to act in parallel or phase with each other such that the discharge and suction cycles of the first and second compressor heads **31**, **51** occur substantially simultaneously. Although certain embodiments of the disclosed compressor modules **100** are hydraulically driven, in other embodiments, other modes of actuating the diaphragms **5** may be implemented. In embodiments, one or more compressor modules **100** may be driven by a crank-slider mechanism (not shown) or other mechanism.

(176) Applicable to any embodiments disclosed herein, the terms “upward” and “downward” are used for convenience in reference to the figures for explaining examples of motion, but are not meant to be limiting. In embodiments, the diaphragm piston **3**, diaphragm **5**, and other components may move in any direction relative to each other, for example left and right, inward and outward, and the like. In embodiments, the diaphragm piston **3** may move perpendicularly or otherwise angled relative to the diaphragm **5** or relative to the actuator piston **126** or other components of the hydraulic drive **110**, so long as actuation movement of the diaphragm piston **3** pressurizes work oil against the diaphragm. In embodiments, the diaphragm piston **3** or intermediate pistons **183** may move in a direction away from or offset from the diaphragm **5**. In other words, by referring to the movement of the piston as the terms “upward” and “downward” with respect to the diaphragm **5** or the compressor head, those terms may be understood as “toward” and “away from,” respectively, or may understood as “pressurizing the work oil” and “depressurizing the work oil,” respectively, or “discharge cycle” and “suction cycle,” respectively.

(177) All of the features disclosed, claimed, and incorporated by reference herein, and all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. Each feature disclosed in this specification may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is an example only of a generic series of equivalent or similar features. Inventive aspects of this disclosure are not restricted to the details of the foregoing embodiments, but rather extend to any novel embodiment, or any novel combination of embodiments, of the features presented in this disclosure, and to any novel embodiment, or any novel combination of embodiments, of the steps of any method or process so disclosed.

(178) Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that any arrangement calculated to achieve the same purpose could be substituted for the specific examples disclosed. This application is intended to cover adaptations or variations of the present subject matter. Therefore, it is intended that the invention be defined by the attached claims and their legal equivalents, as well as the illustrative aspects. The above described embodiments are merely descriptive of its principles and are not to be considered limiting. Further modifications of the invention herein disclosed will occur to those skilled in the respective arts and all such modifications are deemed to be within the scope of the inventive aspects.

Claims

1. A diaphragm compressor system, comprising: a plurality of compressor modules mounted in a stack configuration, each compressor module comprising: a first compressor head and a second compressor head; a diaphragm piston between the first compressor head and the second compressor head; a variable volume region between the diaphragm piston and each of the first compressor head and the second compressor head; each of the first compressor head and second compressor head comprising: a head cavity; and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region; the diaphragm configured to actuate from a first position to a second position during a discharge cycle to pressurize a process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head; wherein the diaphragms of the first and second compressor heads of each compressor module are centered on a compressor axis; a hydraulic drive configured to pressurize work oil and provide the pressurized work oil to the first compressor head and second compressor head, wherein, during the discharge cycle of a compressor head, the hydraulic drive is configured to drive the respective diaphragm piston toward one of the corresponding first compressor head and second compressor head, intensifying the work oil in the respective variable volume region to an intensified pressure, and actuating the diaphragm to the second position; and a clamping mechanism configured to apply a clamping force parallel to the compressor axis, the clamping mechanism comprising a base plate and an end plate on opposing sides of the plurality of compressor modules, the base plate and end plate configured to be compressed, wherein the stack configuration of the plurality of compressor modules is configured such that the first or second compressor head of each compressor module that is not adjacent to the base plate or adjacent to the end plate, contacts a first or second compressor head of an adjacent compressor module.

2. The diaphragm compressor system of claim 1, wherein each compressor head comprises a work oil head support plate and a process gas head support plate; wherein the clamping force of the clamping mechanism is configured to clamp together each work oil head support plate with the respective process gas head support plate for each compressor module of the plurality of compressor modules.

3. The diaphragm compressor system of claim 1, wherein the plurality of compressor modules are in a staged configuration configured to discharge process gas at a first pressure and a second pressure; and wherein the system is configured to provide the discharged process gas at the first pressure from the first compressor head of the first compressor module of the plurality of compressor modules as an inlet supply of process gas to another compressor head of the system.
4. The diaphragm compressor system of claim 3, wherein one or more of the compressor modules of the plurality of compressor modules comprises a bypass check valve configured to bypass process gas past the respective compressor module.
5. The diaphragm compressor system of claim 4, wherein the compressor axis and the actuator axis are coaxial.
6. The diaphragm compressor system of claim 4, wherein the compressor axis and the actuator axis are not coaxial.
7. The diaphragm compressor system of claim 3, wherein each compressor module comprises the first compressor head outputting process gas at the first pressure and the second compressor head outputting process gas at the second pressure; and wherein the system is configured to provide the discharged pressurized process gas from the first compressor head as an inlet supply of process gas to the second compressor head.
8. The diaphragm compressor system of claim 3, wherein the hydraulic drive of each compressor module further comprises an actuator piston defining an actuator axis; and wherein the actuator piston is configured to move along the actuator axis to drive the diaphragm piston.
9. The diaphragm compressor system of claim 3, wherein the diaphragm compressor system is configured to supply work oil from one or more hydraulic power units and from one or more pressure circuits of the plurality of pressure circuits to each hydraulic drive of two or more compressor modules of the plurality of compressor modules.
10. The diaphragm compressor system of claim 1, wherein the plurality of compressor modules comprising four compressor modules.
11. The diaphragm compressor system of claim 10, wherein the four compressor modules are configured to provide four sequential stages of increasing process gas pressurization.
12. The diaphragm compressor system of claim 11, wherein the four compressor modules comprise: a first compressor module configured to output pressurized process gas of at least 50 bar, a second compressor module configured to output pressurized gas of at least 200 bar, a third compressor module configured to output pressurized gas of at least 600 bar, and a fourth compressor module configured to output pressurized gas of at least 800 bar.
13. The diaphragm compressor system of claim 1, wherein the clamping mechanism connecting the base plate and the end plate is at least one of: two or more tie rods and a reactionary frame.
14. The diaphragm compressor system of claim 1, the clamping mechanism comprising: one or more tie rods; and at least one of a plurality of pre-tensioning nuts and a plurality of Belleville spring washers; wherein the clamping mechanism is configured to provide a pre-tension load on at least one of the base plate and the end plate.
15. The diaphragm compressor system of claim 1, wherein the clamping mechanism further comprising a clamp actuator configured to provide a dynamic clamping force to the plurality of compressor modules.
16. The diaphragm compressor system of claim 1, wherein the clamping mechanism comprises: a plurality of tie rods and a plurality of tensioner nuts.
17. The diaphragm compressor system of claim 1, wherein the hydraulic drive comprises: a hydraulic power unit configured to provide a variable-pressure supply of work oil to the hydraulic drive; a plurality of pressure circuits comprising: a first pressure circuit of work oil at a first pressure, and a second pressure circuit of work oil at a second pressure; a first diaphragm piston, wherein a first variable volume region is defined between the first diaphragm piston and the diaphragm of the first compressor head; and a second diaphragm piston, wherein a second variable

volume region is defined between the second diaphragm piston and the diaphragm of the second compressor head.

18. The diaphragm compressor system of claim 1, wherein each compressor module of the plurality of compressor modules is configured to be selectively deactivated; and wherein the diaphragm compressor system is configured to operate the remaining compressor modules of the plurality of compressor modules when a compressor module of the plurality of compressor modules is deactivated.

19. The diaphragm compressor system of claim 1, wherein the clamping mechanism is configured to increase a distance between the base plate and the end plate in response to thermal expansion of one or more compressor modules of the plurality of compressor modules.

20. A diaphragm compressor system, comprising: a plurality of compressor modules mounted in a stack configuration, each compressor module comprising: a first compressor head and a second compressor head; a diaphragm piston between the first compressor head and the second compressor head; a variable volume region between each of the diaphragm piston and the first compressor head and the second compressor head, each of the first compressor head and second compressor head comprising: a head cavity, and a diaphragm mounted in the head cavity and dividing the head cavity into a work oil region and a process gas region, the diaphragm configured to actuate from a first position to a second position during a discharge cycle to pressurize process gas in the process gas region from an inlet pressure to a discharge pressure, and discharge the pressurized process gas through the respective compressor head, wherein the diaphragms of the first and second compressor heads of each compressor module are centered on a compressor axis; a hydraulic drive configured to pressurize work oil and provide the pressurized work oil to the first compressor head and second compressor head, wherein, during the discharge cycle of a compressor head, the hydraulic drive is configured to drive the respective diaphragm piston toward one of the corresponding first diaphragm compressor head and second compressor head, intensifying the work oil in the respective variable volume region to an intensified pressure, and actuating the diaphragm to the second position; and a clamping mechanism configured to apply a clamping force, parallel to the compressor axis, to the first and second compressor head of each compressor module of the plurality of compressor modules, the clamping mechanism comprising a base plate and an end plate on opposing sides of the plurality of compressor modules, the base plate and end plate configured to be compressed, wherein the stack configuration is continuous such that it comprises no gaps along a line parallel to the compressor axis adjacent the first compressor head and second compressor head of each compressor module.
